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Glycobiology-driven therapeutic targeting of glycan-binding proteins: mechanisms, diseases, and clinical translation

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Galectins are glycan-binding proteins that interact with diverse glycoconjugates and protein networks to regulate intracellular and extracellular signaling pathways governing fundamental biological processes. Increasing evidence implicates galectins in the pathogenesis of major human diseases, including cancer, cardiovascular disorders, neurodegeneration, metabolic disorders, and autoimmune conditions, in which they influence angiogenesis, metastasis, immune evasion, fibrosis, and cell fate decisions. Despite the structural similarities of galectins within their conserved carbohydrate recognition domains, individual galectin family members exhibit context-dependent functions and can even exert opposing effects across tissues and disease states, highlighting their complexity. The sixteen mammalian galectins are classified into prototypical, tandem-repeat, and chimeric subtypes, each characterized by distinct binding specificities and thus engaging in the regulation of unique signal transduction mechanisms. Here, we review current understanding of galectin structure, glycan recognition, and the signaling networks they orchestrate in human disease. We highlight their therapeutic potential as both biomarkers and drug targets, and critically evaluate ongoing clinical trials of galectin inhibitors, including modified carbohydrates, pectins, allosteric modulators, and monoclonal antibodies. Finally, we discuss how emerging technologies, such as single-cell glyco-RNA sequencing, CRISPR-based glycoengineering, and targeted protein degradation approaches, are advancing our understanding of galectin biology and accelerating the development of more effective galectin-directed therapeutics.

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INTRODUCTION

While genes and proteins are central determinants of health and diseases, growing evidence indicates that the glycans attached to them represent an additional layer of regulation with important diagnostic and therapeutic potential.[1–3] Recent research in glycobiology has uncovered essential roles for glycans in cell signaling, cell-cell communication, and immune responses.[4–6] Although historically overlooked, glycobiology is now widely recognized as a major contributor to human disease, in particular cancer.[5,7]

Glycobiology is the study of glycans, also known as saccharides, carbohydrates, or sugar chains, including their structure, biosynthesis, and function in living organisms. Glycans are composed of monosaccharide units linked by glycosidic bonds, forming either a linear or branched structure.[8] Through the complex post-translational modification known as glycosylation, glycans are covalently attached to proteins or lipids, to form glycoconjugates, glycolipids, and proteoglycans.[8–10] These glycoconjugates are widely distributed on the cell surface, within the cytoplasm and intracellular organelles, and in the extracellular environment, where they play critical roles in regulating biological processes. Dysregulation of glycosylation is associated with disease states.[8,9] For instance, a hallmark of cancer is aberrant glycosylation, which leads to tumor-derived glycans with altered structures, including tumor-associated carbohydrate antigens (TACAs) and increased branching of N-linked glycans.[11–14]

Central to the biological impact of these altered glycans are glycan-binding proteins (GBPs), which decode glycan information and mediate downstream signaling. Galectins, the major subgroup of GBPs, comprise a conserved family of β-galactoside-binding lectins that regulate a wide range of cellular processes, including apoptosis, angiogenesis, immune suppression, and metastasis.[11,12,15,16] Galectins are uniquely positioned at the interface of the tumor and immune microenvironments, where they sense glycosylation changes and promote tumor progression and therapy resistance.[8,17]

Galectins have emerged as promising therapeutic targets owing to their extracellular accessibility, selective glycan recognition, and immunomodulatory functions.[8,16,18] Several galectin inhibitors have progressed to preclinical and clinical evaluation, highlighting their therapeutic potential in cancer, inflammation, and fibrosis.[18,19] Although best known for their carbohydrate recognition domains (CRDs) that bind β-galactoside-containing glycans, galectins also exert glycan-independent functions, including protein-protein interactions, and modulation of intracellular signal transduction.[20,21] Their pleiotropic and context-dependent roles underscore the need for deeper mechanistic insight to guide the development of effective galectin-targeted therapies.

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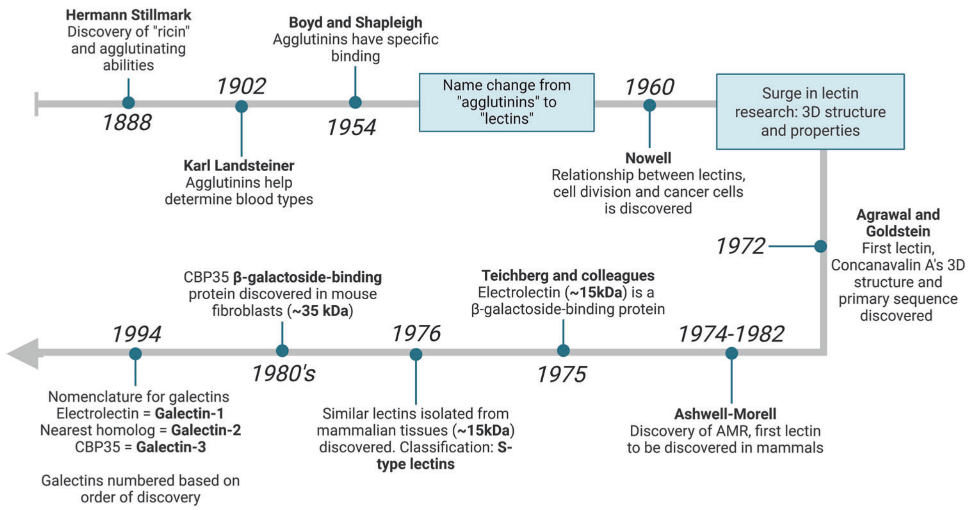

Fig. 1 Timeline leading up to the discovery of GBPs. The timeline includes key research milestones in the development of galectin discovery, starting from 1888 to 1994. BioRender (https://www.biorender.com) was used to create this figure

This review focuses on the galectin family of GBPs, highlighting their structural features, glycan-binding specificity, roles in disease, and emerging therapeutic strategies.

HISTORICAL REVELATION OF RESEARCH AND BREAKTHROUGHS IN GLYCOBIOLOGY AND GBPS

Research in glycobiology, particularly on GBPs and galectins, has advanced substantially over the past decades (Fig. 1). Lectins, a major class of GBPs, are classified into four groups based on structural features, and their CRDs.[22,23] CRDs define lectin specificity by recognizing terminal glycan motifs.[22] Animal lectins include C-type (calcium-dependent), P-type (mannose-6-phosphate-binding), I-type (immunoglobulin [Ig]-like lectins that bind sialic acid—Siglecs), and S-type lectins, which recognize β-galactoside-containing glycans.[22–27] This review focuses on S-type lectins, also known as galectins. The roles and structures of the C-, P-, and I-type lectins have been reviewed elsewhere.[23–27]

The discovery of lectins dates back to 1888, when Hermann Stillmark reported that castor bean seed extracts, containing the toxic protein "ricin," could agglutinate animal erythrocytes together.[22,28–30] Similar hemagglutinating activities were found in other plants, giving rise to the term "agglutinins".[22,29] In 1902, Karl Landsteiner discovered that plant agglutinins could distinguish between different blood types, contributing to the identification of human blood groups A, B and O.[22,29–31] This specificity was further demonstrated in 1954, when Boyd and Shapleigh showed that lima bean extracts agglutinated type A blood but not B or O.[30] Based on these selective binding properties, the term **"lectins"**, derived from the Latin "*legere*", was introduced.[22,28–31]

By 1960, Nowell discovered that the lectin phytohemagglutinin (PHA) stimulates mitogenic activity and cell division.[28] A few years later, Nowell showed that wheat germ agglutinin (WGA) preferentially binds cancer cells. Advances in lectin research accelerated during the 1970s and 1980s with the characterization of their molecular properties and three-dimensional (3D) structures. In 1972, Agrawal and Goldstein introduced affinity chromatography for lectin purification, enabling the determination of their primary sequences and 3D structures, including that of Concanavalin A. Advances in purification methods and the discovery of recombinant DNA technology further accelerated this field.[28]

A major turning point occurred between 1974 and 1982, with the identification of Ashwell-Morell asialoglycoprotein receptor (AMR) in hepatocytes, the first mammalian lectin.[31–33] Shortly after, Teichberg and colleagues discovered a ~15 kDa lectin from the organs of the electric eel, termed "electrolectin," which hemagglutinated trypsinized rabbit erythrocytes.[31,34] Similar lectins were subsequently identified in mammals, including chicken muscle, calf heart, and lungs.[31] It was determined that these proteins had free cysteine residues and relied on their sulfhydryl groups, thus classified as "S-type lectins". By the 1980's, a 35 kDa β-galactoside-binding protein, CBP35, was identified in mouse fibroblasts. In 1994, standardized nomenclature designated electrolectin as Galectin-1 (Gal-1), its homolog Galectin-2 (Gal-2), and CBP35 as Galectin-3 (Gal-3), with subsequent family members numbered according to their discovery.[31]

TYPES OF GALECTINS: STRUCTURAL AND FUNCTIONAL OVERVIEW

The S-type lectin subgroup, known as galectins, regulates diverse biological processes including signal transduction, cellular cross-talk, and immune responses.[35,36] Galectins are expressed in epithelial, immune, neuronal, and endothelial cells, and are often indicators of cellular stress or injury.[35–37] To date, sixteen mammalian galectins have been identified and are classified into three structural groups. Prototypical galectins -1, -2, -5, -7, -10, -11, -13, -14, -15, and -16 contain a single CRD. Tandem-repeat galectins -4, -6, -8, -9, and -12 possess two homologous CRDs, connected by a peptide linker.[31,36,38] Chimeric galectins, represented solely by Gal-3, possess a single CRD and an N-terminal domain rich in proline, glycine, and tyrosine, which facilitates oligomerization[31,36,38,39] (Fig. 2). Prototypical galectins exist as monomers or non-covalent homodimers, tandem-repeat galectins form heterodimers, and Gal-3 can assemble into pentamers.[36,40–42]

All galectins share conserved β-galactoside specificity mediated by their CRD.[31,36] The functional domain of the CRD (~130 amino acids) forms a β-sandwich composed of two anti-parallel β-sheets that generate a glycan-binding groove. The convex face is comprised of five antiparallel strands called F1–F5, while the concave face contains six antiparallel strands called S1–S6. Conserved residues in strands S4–S6 mediate β-galactoside binding, while strands S1–S3 facilitate interactions with larger oligosaccharide binding[36] (Fig. 3).

Through the course of evolution, galectins have undergone structural diversification. Initially containing a single CRD, gene duplication led to the emergence of galectins with two CRDs positioned at the N- and C-termini.[31,36] These domains differ in

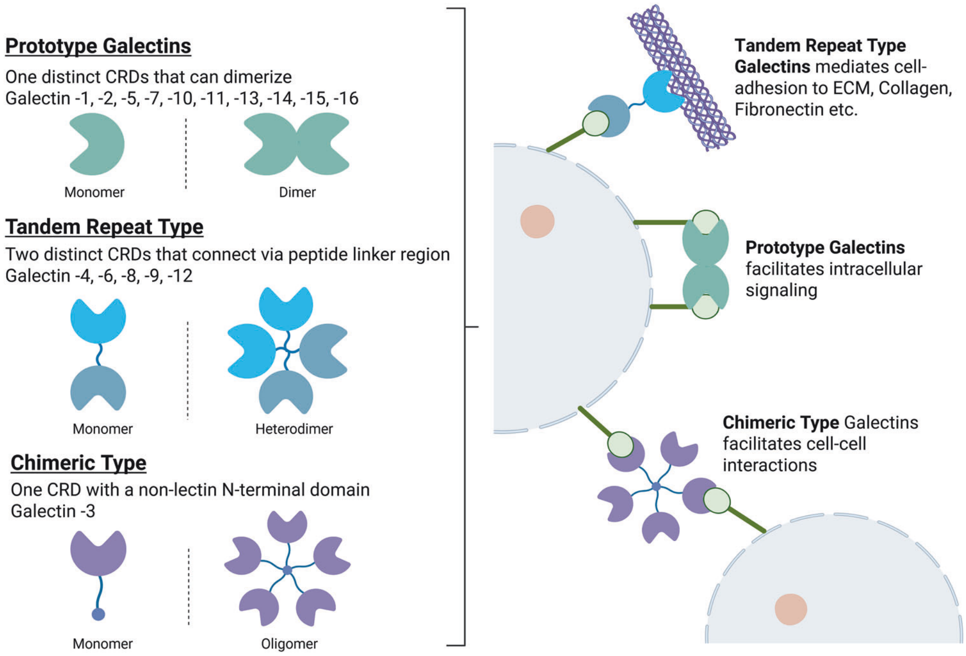

Fig. 2 Classification of galectins based on their structure and function. Prototype galectins include galectins-1, -2, -5, -7, -10, -11, -13, -14, -15, -16; Tandem-repeat type galectins include galectins -4, -6, -8, -9, -12; The chimeric type galectin group consists solely of Gal-3. BioRender (https://www.biorender.com) was used to create this figure

exon-intron organization and are classified as F4-CRD (N-terminal) and F3-CRD (C-terminal). Single-CRD galectins belong to either the F4-CRD (galectin-7, -10, -11, -13, -14, -15) or the F3-CRD (galectin-1, -2, -3, -5), whereas tandem-repeat galectins (galectin-4, -6, -8, -9, -12) contain both domains, enabling dual glycan recognition.[31,36,43,44]

Galectin-1 (Gal-1)

Gal-1, a prototype galectin encoded by the *LGALS1* gene on chromosome 22q13.1[36,45] exists as a 14.5 kDa monomer,[46] but can also dimerize through non-covalent hydrophobic interactions.[21,40] Each monomer has a single CRD composed of five F-strands (F1–F5) and six S-strands (S1–S6a/b) forming an anti-parallel β-sandwich.[21,36] In its dimer form, two monomeric units are arranged such that their CRDs are positioned on opposing ends of the quaternary structure, creating a 22-strand anti-parallel β-sandwich,[47,48] enabling multivalent binding to glycosylated ligands, cell-surface lattice formation, and downstream signaling.[40,47] Although the homodimer is required for efficient signaling, the monomer retains limited glycan binding activity with lower affinity.[21,49]

Gal-1 function is tightly regulated by redox state.[49] Each monomer contains six cysteine residues (Cys2, Cys16, Cys42, Cys60, Cys88 and Cys130), which makes it highly susceptible to oxidation.[36,48–50] Oxidation promotes the formation of three intramolecular disulfide bonds, abolishes lectin binding, and prevents dimerization.[36,49–52] By contrast, β-galactoside ligands such as lactose or *N*-Acetyllactosamine (LacNAc) stabilize the reduced state and favor dimerization.[50,52] Gal-1 activates signaling pathways, including integrin α6β4, Notch1/Jagged2, and vascular endothelial growth factor receptor 2 (VEGFR2), thereby promoting cancer cell proliferation, angiogenesis, and tumor progression.[53,54]

Gal-2

Gal-2 is a prototypical galectin encoded by the *LGALS2* gene, located on chromosome 22q13.1 (the same as *LGALS1*), ~50 kb away on the opposite strand of *LGALS1*. Structurally, Gal-2 shares 43% homology with Gal-1. The protein has a molecular weight of 14 kDa and exists in both monomeric and non-covalent homodimer forms.[36,55] Gal-2 is primarily found in the gastrointestinal tract, although its presence in the placenta and cardiovascular system are also reported.[55,56] As a monomer, its CRD is divided into four subsites (A–D).[55] Subsite C of the CRD is a highly conserved region composed of six amino acids and is the primary binding site for β-galactoside. Subsites A, B, and D can bind flanking units of the primary β-galactoside, providing a more stable and specific binding. Within its CRD structure, Gal-2's sugar-binding sites are on S4–S6. Interestingly, Gal-2 can form non-covalent homodimers when in solution on strands F1 and S1, in which four parallel hydrogen bonds are formed in each β-sheet.[55]

Gal-2 has been implicated in gastrointestinal disease, cancer, myocardial infarction, and atherosclerosis.[55–57] Gal-2 binding partners, such as monosialotetrahexosylganglioside (GM1 ganglioside) in neuroblastoma, and mucin 1 (MUC1) in colon and breast cancer, are associated with increased risk of metastasis.[55,58] Additionally, its downstream effects in interacting with cytokine lymphotoxin-a (LTA) increase its detrimental effects on myocardial infarction and atherosclerosis.[36,55,57]

Gal-3

Gal-3, encoded by the *LGALS3* gene, is the only galectin belonging to the chimeric-type group.[59] It is located on chromosome 14, positions q21-q22, with a molecular weight of approximately 35 kDa.[36] Gal-3 possesses a proline-rich N-terminal domain that takes part in oligomerization, secretion, and nuclear translocation.[36,39] Its C-terminal domain holds the CRD, which binds LacNAc residues, facilitating interactions with glycoconjugates containing *N*-acetylglucosamine.[36,60] Within the CRD domain, a sequence of four amino acids, Asn-Trp-Gly-Arg (residues 180–183), makes up the NWGR anti-apoptotic motif.[36,61]

Gal-3 has been widely investigated as a prognostic biomarker in cardiovascular, renal, autoimmune, metabolic, neurodegenerative, and malignant diseases.[62–66] Gal-3 is shown to regulate metastasis in different human malignancies, including melanoma, non-Hodgkin's lymphoma, and lung, breast, ovarian, and colorectal cancers.[67–70]

Galectin-4 (Gal-4)

Gal-4, a tandem-repeat galectin, is encoded by the *LGALS4* gene located at chromosome 19, position q13.1-2.[36] It is known for

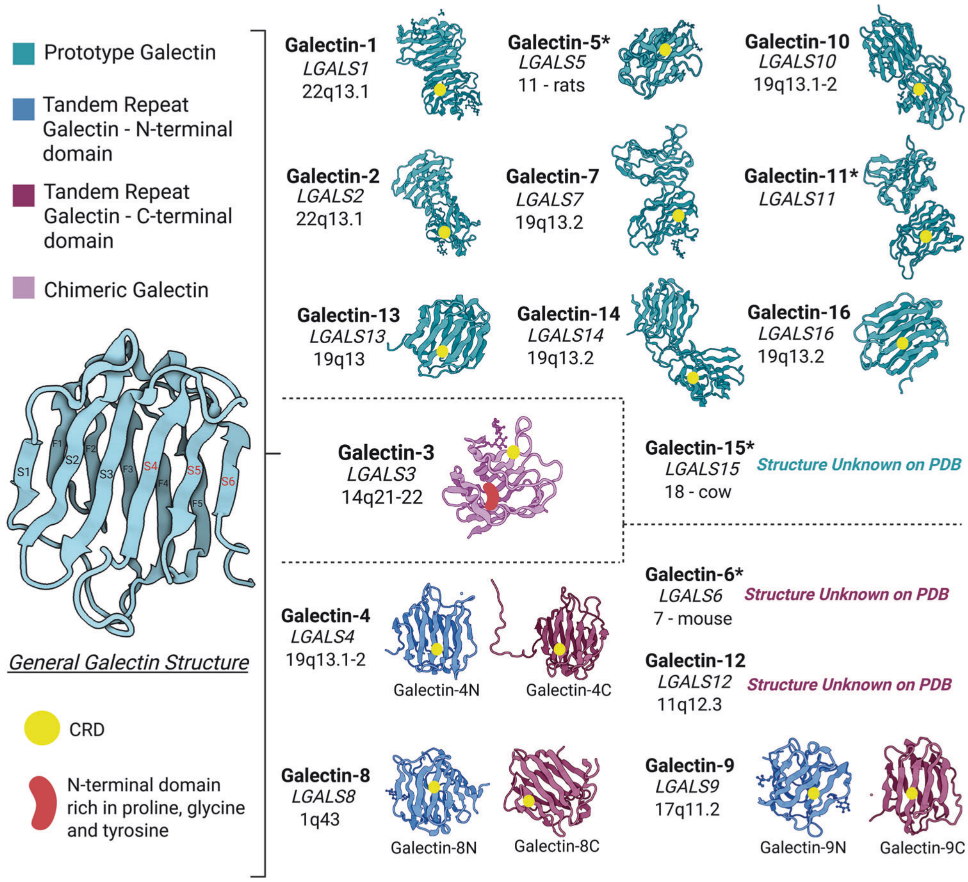

Fig. 3 Structural comparison of Gal-1 to galectin-16. General galectin structure indicates the CRD structure made up of strands S1–S6 on the concave face, with strands F1–F5 on the convex face. S4–S6 are crucial for β-galactoside binding. Prototypical galectins are shown to possess one CRD and are in both monomeric and dimeric structures; Tandem-repeat galectins are depicted in their N-terminal domain CRD, and C-terminal domain CRD; Chimeric galectin is indicated by having a single CRD and an N-terminal domain rich in proline, glycine, and tyrosine. *Indicates galectins that are not found in humans. Protein structures were retrieved from the PDB; the CRD is shown for illustrative purposes and may not correspond to the exact structural location. Gal-1 (PDB ID: 1GZW), Gal-2 (PDB ID: 5DG2), Gal-3 (PDB ID: 4XBN), Gal-4N (PDB ID: 4XZP); Gal-4C (PDB ID: 1X50), Gal-5 (PDB ID: 5JP5), Gal-7 (PDB ID: 4GAL), Gal-8N (PDB ID: 5GZD); Gal-8C (PDB ID: 8HL9), Gal-9N (PDB ID: 2EAK); Gal-9C (PDB ID: 3NV1), Gal-10 (PDB ID: 5XRG), Gal-11 (PDB ID: 6N3R), Gal-13 (PDB ID: 6KJW), Gal-14 (PDB ID: 6K2Y), Gal-16 (PDB ID: 6LJP). BioRender (https://www.biorender.com) was used to create this figure

having two CRDs, the first located at the N-terminal domain (Gal-4N) and the second at the C-terminal domain (Gal-4C), both connected by a linker-peptide bridge.[36] The linker-peptide bridge is proline- and glycine-rich and contains approximately 30 amino acids, thereby providing structural flexibility.[71] An extended region encompassing amino acids 170-173 forms a short alpha-helix that contributes to structural stability.[72] Together, these regions allow the two CRDs to adopt conformations that facilitate protein-protein interactions and conformational changes, while maintaining ligand-binding specificity.[71,72] Due to this specificity, each CRD preferentially binds distinct ligand groups.[36] The linker-peptide bridge has been reported to act as a flexible hinge, that may influence the interaction and crosstalk between Gal-4N and Gal-4C.[72]

Gal-4 is naturally expressed in healthy gastrointestinal tissue; however, its expression pattern differs in diseased tissue. Its structural adaptability appears to contribute to diverse biological responses and has been implicated in the modulation of signaling pathways associated with tumorigenesis.[71,72] For example, Gal-4 is detected in stomach and liver cancer and has been proposed as a promising drug target.[36,73]

Galectin-5 (Gal-5) and Galectin-6 (Gal-6)

Gal-5 and Gal-6 are both known to be expressed in rodents[31] and are, therefore, not discussed in detail in this review. Gal-5, a prototypical galectin, is encoded by the *LGALS5* gene and is located on chromosome 11 in rats and chromosome 17q11 in the human syntenic region.[74] Originally extracted from rat lung tissue, Gal-5 was identified through sequencing using a rat reticulocyte cDNA library and demonstrated more than 80% homology with the C-terminal CRD sequence of rat Galectin-9 (Gal-9). Gal-5 exists as a monomer in solution but can also form oligomers under non-denaturing conditions.[36,75]

Gal-6 is a tandem-repeat galectin discovered in the mouse gastrointestinal tract.[76,77] Gal-6 is encoded by the *LGALS6* gene, located on mouse chromosome 7.[76] Using Gal-4 cDNA cloning derived from mouse colon, 83% amino acid homology with Gal-6 was demonstrated.[77] A recent study links Gal-6 expression to altered skin immunity in mice lacking key epidermal barrier proteins, with effects influenced by skin microbiota.[78]

Galectin-7 (Gal-7)

Gal-7 is a prototypical galectin, encoded by the *LGALS7* gene[79] on chromosome 19q13.2, and has a molecular weight of 15 kDa.[36,80,81] Gal-7 exists as both a monomer and a homodimer and has high affinity for internal and terminal LacNAc. A conformational change caused by a serine substitution of arginine at position 74 (R74S) abolishes its carbohydrate-binding activity without disrupting dimerization.[81]

Initially identified as a keratinocyte-specific marker, Gal-7 is expressed throughout epidermis as well as in stratified epithelial tissues, including the tongue, cornea, esophagus, rectal mucosa, and Hassall's corpuscles of the thymus.[79,80,82] Gal-7 has also been implicated in skin disease and carcinogenesis and is being explored as a target in non-melanoma skin cancer.[79,82–85]

Galectin-8 (Gal-8)

Gal-8 is a tandem-repeat galectin encoded by the *LGALS8* gene on chromosome 1q43, with a molecular weight of approximately 34 kDa.[86,87] Structurally, Gal-8 contains two CRDs connected by a linker region and likely forms homodimers through its N-terminal domain.[88] Alternative splicing of *LGALS8* generates four isoforms that differ in peptide linker from 24 to 74 amino acids, thereby altering CRD spacing, ligand affinity, and function.[89]

Gal-8 has been studied in immunity, microbial infections, and cancer.[87] In normal tissues, Gal-8 supports innate immune response by inducing pro-inflammatory cytokines, whereas in cancer, it promotes immune evasion, T cell dysfunction, and metastasis.[87,90] Gal-8 is being explored as a druggable target in breast, prostate, and lung cancers.[87,89]

Gal-9

Gal-9 is encoded by the *LGALS9* gene located on chromosome 17q11.2 with a molecular weight of 34–39 kDa.[91,92] Characterized as a tandem-repeat type galectin, Gal-9 has two CRDs known as the N-CRD and the C-CRD, sharing 35-39% sequence homology with each other.[93–95] Both CRDs possess a β-sandwich with two sets of antiparallel β-strands called S1–S6 and F1–F5. Within this β-sandwich is a short α-helix. Each CRD differs in terms of functional roles and binding affinities due to differences in strands S4–S6.[94,95] Interestingly, the linker connecting both CRDs contains protease-sensitive sites, which allows for Gal-9 to be cleaved.[96,97]

Gal-9 expression is either increased or decreased depending on tumor type, leading to tumor-promoting or tumor-suppressive effects.[91] In acute myeloid leukemia (AML), Gal-9 promotes leukemic stem cell survival and AML progression through a T cell immunoglobulin and mucin domain-3 (TIM-3)-dependent autocrine loop.[98] By contrast, in colorectal cancer, high Gal-9 expression is associated with improved prognosis and survival, in part through induction of apoptosis.[97] In addition, Gal-9 N-CRD is involved in activating dendritic cells (DCs), while Gal-9 C-CRD contributes to T cell death and promotes anti-proliferation.[93,99] Therefore, Gal-9 may serve as a suitable biomarker for predicting disease progression and severity in a context-dependent manner.[92]

Galectin-10 (Gal-10)

Gal-10, a prototypical member with a molecular weight of approximately 16 kDa, is encoded by the *LGALS10* gene on chromosome 19q13, reported as 19q13.1 in early mapping studies[100] and 19q13.2 in current genomic databases (NCBI Gene ID: 1178).[36,100] Researchers have shown that Gal-10 dimerizes in a manner distinct from other prototypical galectins.[101] Its two CRDs face one another in an S-to-S face configuration, unlike other galectins, where their S face is exposed, thus changing their ligand binding specificity.[36,102] This arrangement is mediated by an α-helix, linking the S3-F2 β-strands to opposite monomers.[102,103] In this configuration, Glu33 in one monomer blocks the carbohydrate-binding site of the other. The mutation of Glu33 to alanine (E33A), shifts Gal-10 from a dimeric to a monomeric state and restores lactose binding. Another key residue is Trp72. Mutation of Trp to alanine (W72A), prevents dimerization and increases both agglutination and lactose binding.[101,103]

Gal-10 is abundantly expressed in human eosinophils,[104–106], and is proposed as a biomarker for diverse diseases, including asthma, eosinophilic esophagitis, rhinitis, and sinusitis.[104] Although Gal-10 appears to play a role in eosinophilic inflammation in different diseases, further research is required to understand Gal-10's specific mechanisms in signal transduction.

Galectin-11 (Gal-11) and Galectin-15 (Gal-15)

Gal-11 and Gal-15 are prototypical galectins found in ruminants,[107,108], and are therefore of limited relevance to human biology. Gal-11, encoded by *LGALS11*, is expressed in the nuclei and cytoplasm of gastrointestinal and epithelial cells, and contributes to host defense against gastrointestinal nematodes in animals.[107,109] Gal-11 exists as two isoforms that differ by 10.9% and have anti-parasitic activities.[107,109] Gal-11 is, therefore, of interest as a determinant of resistance to gastrointestinal parasite infection in animals.[110]

Gal-15, encoded by *LGALS15*, located on cow chromosome 18, is expressed in the uterus and gastrointestinal tissues of the Artiodactyla species, including sheep, goat, and cattle.[108,111] It contributes to the control of inflammation and eosinophil infiltration during infections and also supports trophoblast attachment during the peri-implantation period.[108,112] Although there is no direct human orthologue, Gal-15 is most similar in structure and amino acid composition to human Gal-10 and -13.[108,113]

Galectin-12 (Gal-12)

Gal-12 is encoded by the *LGALS12* gene, on chromosome 11q12.3 (NCBI Gene ID: 85329), and has a molecular weight of approximately 37 kDa.[36,114] Being a part of the tandem-repeat family, it possesses two CRDs connected by a peptide linker chain.[115] Although its N-terminal CRD has a sequence similarity of 32–43% to other galectins, the C-terminal CRD has only a 23–28% similarity, making it more difficult to bind specific β-galactosides.[114–116]

Gal-12 is reported to play key roles in cellular growth, differentiation, and apoptosis, but also appears to be involved in tumor suppression.[117,118] Gal-12 overexpression induces cancer cell cycle arrest at the G1 phase and suppresses proliferation.[73,117] Gal-12 mRNA is suggested to contain AU-rich untranslated regions, associated with transcript instability, and a weak Kozak start codon, reducing translation efficiency.[116,117] These features may contribute to low Gal-12 protein expression in tumors such as colon cancer, prostate cancer, and acute lymphatic leukemia. Further work is needed to understand Gal-12's mechanism in diseases.

Galectin-13 (Gal-13) and Galectin-14 (Gal-14)

Gal-13 and Gal-14 are placental-associated galectins, that contribute to immune regulation during pregnancy by inducing apoptosis in activated T lymphocytes.[119,120] Gal-13, encoded by *LGALS13*, on chromosome 19q13 has a molecular weight of approximately 16 kDa. Unlike most galectins, Gal-13 does not bind β-galactosides.[121] Due to its unique amino acid sequence, Gal-13's dimeric structure is held together by disulfide bonds between residues Cys136 and Cys138 at its C-terminal. Being redox-active, these disulfide bonds are reversible and can be created as easily as broken, thus affecting Gal-13's function and structure.[122] Gal-14 is located on chromosome 19 but at a slightly different position, q13.2.[123] Encoded by the *LGALS14* gene with a molecular weight of 16-18 kDa, Gal-14 exists as a monomer or a dimer.[123,124] β-strands S5 and S6 from one monomer interact with another monomer to facilitate dimerization.[123]

Both Gal-13 and -14 are secreted from the placenta into the maternal blood, particularly during the first trimester of pregnancy; however, if a miscarriage occurs, Gal-13 and -14 expression is significantly declined.[119,125] Reduced expression, particularly of Gal-13, is also associated with severe preeclampsia, consistent with important roles in establishing maternal-fetal immune tolerance.[119–121,126]

Galectin-16 (Gal-16)

Gal-16 is a 16 kDa protein, encoded by the *LGALS16* gene on chromosome 19q13.2. This member is similarly believed to aid in

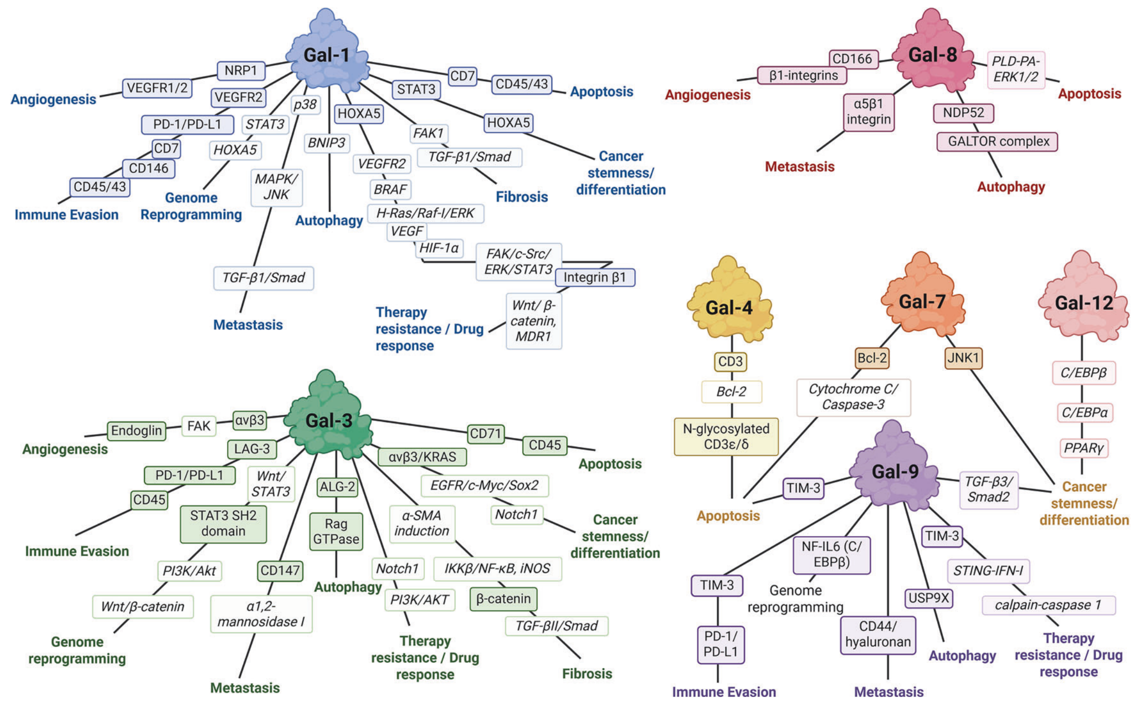

Fig. 4 Functional partners of galectins-1, -3, -4, -7, -8, -9, -12. Each interaction signifies the end effect of angiogenesis, immune evasion, genome reprogramming, metastasis, autophagy, fibrosis, cancer stemness/differentiation, therapy resistance/drug response, or apoptosis. Direct binding partners are indicated by the darker color, while downstream effector partners are shown in the lighter gradient. BioRender (https://www.biorender.com) was used to create this figure

placental development.[127] The only protein structure studied to date is the recombinant monomeric Gal-16, which has the common β-sandwich CRD structure.[127] Based on its structural classification as a single CRD, it is considered a prototypical galectin.[38,128] Further research is required to fully understand Gal-16's mechanisms of action in health and disease.

Intracellular vs extracellular roles

Although synthesized in the cytosol, galectins are detected both intracellularly and extracellularly.[129,130] Unlike most secreted proteins, galectins lack an N-terminal signal peptide and therefore do not enter the conventional endoplasmic reticulum–Golgi secretory pathway.[129,131,132] Instead, they are secreted through unconventional non-classical pathways,[132,133] which are classified into four types[134]: 1) direct translocation of proteins across the plasma membrane (type I); 2) ATP-binding cassettes transporter-mediated export (type II); 3) organelle-associated secretion involving compartments such as lysosomes, endosomes, extracellular vesicles, and microvesicles (type III); and Golgi bypass by transmembrane proteins (type IV). Detailed mechanisms of secretion have been reviewed elsewhere.[129,134] Although galectins may use multiple unconventional secretion routes, export via microvesicles and exosomes appears to be an important mechanism, often involving membrane blebbing and release into the extracellular space.[130,135] At the extracellular space, they bind diverse glycoconjugates and glycosylated receptors, enabling signaling cascades that affect cell growth, immune response, and cellular processes.[37] At the intracellular level, Gal-1 and -3, also have major roles in cell fate because of their ability to interact with non-glycosylated proteins to mediate pre-mRNA splicing, stabilization, and apoptosis.[34,136,137]

Collectively, human galectins comprise single-CRD prototypical members (Gal-1, -2, -7, -10, -13, -14, and -16), tandem-repeat galectins (Gal-4, -8, -9, and -12), and the chimeric Gal-3. Their structural features, including CRD topology, dimerization states, and linker flexibility, highlight their ability to interact with specific glycoconjugates and mediate multivalent binding. Some galectins (Gal-5, Gal-6, Gal-11, and Gal-15), which are not found in humans but are expressed primarily in rodents or ruminants, are not considered further in this human-focused review.

SIGNALING PATHWAYS AND REGULATORY MECHANISMS INVOLVED IN GALECTINS

Mounting research in the field of glycobiology has revealed that galectins play central roles in disrupting immune responses, transcriptional reprogramming, and physiological processes.[138,139] Each galectin has distinct regulatory mechanisms; however, some are better characterized than others (Fig. 4). For this reason, here we focus on galectin family members that are well characterized to date.

Immune modulation

Cancer remains a leading cause of death worldwide, in part due to immune escape. Although immunotherapies have improved outcomes in many cancers, a substantial proportion of patients do not respond to these treatments. Antitumor immunity relies on a coordinated immune cycle. Cancer cells present tumor-associated antigens on their major histocompatibility complex I (MHC-I) molecule on their surface.[140] Once a cancer cell dies, these antigens are captured by antigen-presenting cells (APC) such as DCs, which migrate to draining lymph nodes.[141] There, DCs present the antigens to naive T cells using MHC-I and major histocompatibility complex II (MHC-II) molecules, activating $CD8^+$ T cells (cytotoxic T lymphocytes) and $CD4^+$ T cells (helper T cells), respectively.[140–143] Activated effector T cells, subsequently enter the bloodstream to infiltrate the tumor microenvironment,[141] where recognition of tumor antigens via T cell receptor–MHC interactions triggers cytotoxic responses through perforin and granzyme.[141,144] Natural killer (NK) cells also contribute to tumor clearance by producing inflammatory cytokines and chemokines. Some activated T cells differentiate into memory cells that provide long-term immune surveillance.[145,146]

Immune tolerance is maintained in part by FOXp3$^+$ regulatory T cells (Tregs), which suppress excessive immune activation and prevent autoimmunity.[147,148] The presence of excess Tregs in the tumor microenvironment suppresses CD8$^+$ T cells, NK cells, and DCs, promoting tumor growth.[148–151] Dysregulation of any of these key steps in the cancer immunity cycle facilitates immune evasion.[141] Galectins have emerged as important mediators of these immunosuppressive processes.[152]

Galectins regulate antitumor immunity through several mechanisms, including induction of T cell apoptosis, suppression of T cell activation, promotion of anti-inflammatory T-helper 2 (Th2) responses, expansion of Foxp3$^+$ Treg cells, generation of tolerogenic DCs, inhibition of NK cell activity, and polarization of macrophages towards an immunosuppressive M2 phenotype.[152] In addition, galectins interact with immune checkpoint receptors, including TIM-3, lymphocyte activation gene-3 (LAG-3), programmed cell death protein-1 (PD-1), leukocyte immunoglobulin-like receptor subfamily B3 (LILRB3), and the leukocyte immunoglobulin-like receptor subfamily B4 (LILRB4), to inhibit regulatory immune pathways.[153]

Among galectins, Gal-1 has been extensively studied in immune evasion.[154] In various cancers, including melanoma, head and neck squamous cell carcinoma (HNSCC), neuroblastoma, glioblastoma, and lung cancer, Gal-1's elevated expression is correlated with poor survival, and genetic deletion of *LGALS1*, the gene encoding Gal-1, improves survival in experimental models.[154,155] Gal-1 preferentially binds lactosamine sequences (Galβ1,4GlcNAc) including glycosylated receptors, CD45, CD7, CD43, CD146, VEGFR2, thereby inducing T cell death.[153,156,157] Gal-1 is also able to selectively induce death of Th1 and Th17 cells, while increasing the number of CD4$^+$ and CD8$^+$ regulatory T cells, thus favoring a Th2 biased environment.[139,153] In addition, Gal-1 can upregulate the expression of programmed death-ligand 1 (PD-L1) and prevent T cell infiltration into tumors, resulting in immune escape, metastasis, and angiogenesis.[158,159] Gal-3 similarly modulates immune checkpoint signaling by interfering with PD-1/PD-L1 interactions. Even in the presence of checkpoint inhibitors such as pembrolizumab or atezolizumab, Gal-3 can maintain PD-1/PD-L1 signaling and promote immune escape in metastatic melanoma.[160,161] In triple-negative breast cancer (TNBC), Gal-3 forms a complex with CD45 that increases Tregs, promoting immune intolerance; while in pancreatic cancer, it binds to lymphocyte-activation gene 3 (LAG-3) to induce CD8$^+$ T cell dysfunction.[153,162]

Gal-9 also contributes to immune evasion through its interactions with TIM-3.[153,163] In AML, Gal-9/TIM-3 complex suppresses NK cell-mediated cytotoxicity and increases myeloid-derived suppressor cells (MDSCs), promoting immune evasion.[153] Similar to Gal-1 and Gal-3, Gal-9 can bind PD-1, forming a complex lattice that promotes persistence of PD-1$^+$ TIM-3$^+$ T cell survival.[153,164] In fact, this signaling was shown to activate the NF-κB and β-catenin pathway to enable leukemic stem cell self-renewal.[98] Additional immunosuppressive receptors found including LILRB3 and LILRB4, bind Gal-4, Gal-7, and Gal-8 respectively, to further inhibit immune activation[90,153,165] (Table 1).

Overall, galectins are central regulators of tumor immune evasion. By inducing apoptosis of effector T cells, expanding regulatory T cell populations, and interacting with immune checkpoints such as TIM-3, PD-1, and LAG-3, galectins suppress antitumor immunity and facilitate tumor progression. Understanding these interactions and their specific glycan ligands will be critical for developing therapeutic strategies that target galectin-mediated immunosuppression.

Angiogenesis and vascular remodeling

Angiogenesis, the formation of new blood vessels, is important in several diseases, including cancer and cardiovascular diseases (CVDs). Angiogenesis helps restore blood flow and revascularization in ischemic heart disease; however, excessive angiogenesis can worsen tumor growth and accelerate atherosclerosis due to pro-angiogenic factors, including VEGF and fibroblast growth factor (FGF)[166] (Table 1).

Galectins contribute significantly to tumor angiogenesis by modulating endothelial cell function and ECM interactions. Gal-1 binds neuropilin-1 (NRP1) via its CRD, thereby promoting the formation of NRP1/VEGFR2 heterodimers on endothelial cells. This triggers extracellular signal-regulated kinases 1 and 2 (ERK1/2) phosphorylation and tubular morphogenesis, even in low VEGF environments.[53,167] Gal-1 has also been linked to VEGFR1/NRP1 complex formation, which not only promotes angiogenesis but also contributes to vascular permeability in tumors.[168] Compelling evidence from Croci et al. demonstrated that altered glycosylation of VEGFR2 enables Gal-1 to activate VEGF-like signaling in murine tumors that are resistant to anti-VEGF therapies. These findings identify glycosylation as a crucial determinant of endothelial cell sensitivity to galectin-mediated angiogenic signaling.[167] This further highlights the importance of endothelial glycosylation patterns in shaping tumor vascular responses to galectins and anti-angiogenic therapies.[169,170]

Gal-3 similarly facilitates VEGFR signaling and enhances tumor angiogenesis by clustering integrins such as αvβ3, resulting in activation of the focal adhesion kinase (FAK) pathway.[171] It also interacts with endoglin, a TGF-β co-receptor involved in angiogenesis, and stabilizes the JAG1/Notch-1 signaling in endothelial cells, further linking its function to vascular regulation.[172] Proteolytic cleavage of Gal-3 by matrix metalloproteases (MMP) MMP-2/-9 enhances endothelial migration, FAK activation, and angiogenic responses, underscoring the regulatory role of its N-terminal domain in tumor angiogenesis.[171,173]

Beyond Gal-1 and Gal-3, Gal-8 has emerged as an angiogenic modulator. It interacts with endothelial CD166 (ALCAM) and various integrins to stimulate migration, capillary-tube formation, and vascular permeability in vitro and in vivo.[174,175] Elevated Gal-8 serum levels correlate with enhanced angiogenesis in breast and colorectal cancers, suggesting its potential as a biomarker and therapeutic target.[58,176]

The roles of other galectins, such as Gal-9 and Gal-12, in tumor angiogenesis are less clear. Gal-9 can act as a chemoattractant for endothelial cells, modulating migration and tube formation in a concentration- and isoform-dependent manner, but its effects on VEGFR signaling remain unknown.[177] Gal-12, though not expressed by endothelial cells, can induce angiogenic responses via modulation of fucosylated glycans under hypoxic conditions, a pathway potentially relevant to the tumor microenvironment.[178]

Taken together, the literature has established a multifaceted angiogenic program orchestrated by galectins, converging on VEGFR modulation, integrin clustering, and TGF-β signaling crosstalk, ultimately sustaining tumor vascular plasticity and resistance to anti-angiogenic therapies.

Metastasis and cell migration

Galectins play pivotal roles in cancer metastasis by linking extracellular glycan recognition to intracellular signaling pathways that govern cytoskeletal remodeling, cell adhesion, and motility. Gal-1 promotes epithelial-mesenchymal transition (EMT) and metastatic potential in gastric cancer via β1 integrin-dependent activation of TGF-β1/Smad signaling.[179,180] It also activates MAPK pathways, including JNK and p38, increasing MMP secretion and facilitating migration in ovarian cancer.[180]

Gal-3 similarly enhances cell motility by activating Rac1 GTPase and inducing FAK phosphorylation (Y397), key features of integrin-driven migration.[181] Additionally, Gal-3 promotes CD147 clustering at the leading edge of corneal epithelial cells, stimulating MMP-9 expression and cell migration. This process depends on α1,2-mannosidase I activity and the CD147 glycan interaction.[182,183] Gal-8 also promotes EMT by clustering α5β1 integrin

Table 1. Summary of galectin binding partners and downstream effectors

Galectin	Interacting partners*	Promote disease yes/no	Biological process/condition	Effect	Ref.
Immune modulation					
Gal-1	CD45, CD7, CD43, CD146, VEGFR2, PD-1/PD-L1	Yes	Melanoma, HNSCC, neuroblastoma, glioblastoma, lung cancer	T cell death, Th2 anti-inflammation, prevent migration, immune escape, metastasis, angiogenesis	153,156–159
Gal-3	PD-1/PD-L1, CD45, LAG-3	Yes	MM, TNBC, pancreatic cancer	Immune escape, increase Tregs, immune intolerance, $CD8^+$ T cell dysfunction	153,160–162
Gal-9	TIM-3, PD-1/PD-L1	Yes	AML	Immune evasion, CSC self-renewal	153,164
Angiogenesis and vascular remodeling					
Gal-1	NRP1, VEGFR1/2	Yes	Tumor angiogenesis	Vascular remodeling	53,167,168
Gal-3	αvβ3, Endoglin, *FAK*	Yes	Tumor angiogenesis	Tube formation, tumor growth	171,172,357
Gal-8	CD166, β1-integrins	Yes	Breast cancer, colorectal cancer	Tube formation, vascular permeability	174–176
Metastasis and cell migration					
Gal-1	*TGF-β1/Smad, MAPK/JNK, p38*	Yes	Cancer metastasis/associated fibroblasts	EMT, increase MMP secretion, CSC migration	180
Gal-3	*a1,2-mannosidase I*, CD147	Yes	Cancer metastasis	Increase MMP secretion, CSC migration	182,183
Gal-8	α5β1 integrin	Yes	Cancer metastasis	Activation of EGFR leading to EMT	184
Gal-9	CD44/hyaluronan	Yes	Cancer metastasis	Tumor cell adhesion and migration	185
Apoptosis and cell survival					
Gal-1	CD45, CD43, CD7	No	Apoptosis and cell survival	Pro-apoptosis	156,189–192
Gal-3	CD45, CD71	Yes	Apoptosis and cell survival	Block apoptosis	36,194,195,199
Gal-4	CD3	No	Apoptosis and cell survival	Induce apoptosis	200
	Downregulates Bcl-2	Yes	Colorectal cancer	Enhance apoptosis	201
	N-glycosylated CD3ε/δ	Yes	Pancreatic cancer	Induce T cell apoptosis, immune evasion, tumor progression	200
Gal-7	*Cytochrome C/Caspase-3*, Bcl-2	No	Apoptosis and cell survival	Induces apoptosis	202,358
Gal-8	*PLD-PA-ERK1/2*	No	Apoptosis and cell survival	Induces apoptosis	203,270
Gal-9	TIM-3	Yes	Apoptosis and cell survival	Immune tolerance, promotes survival in leukemic stem cells	204
Autophagy and metabolic reprogramming					
Gal-1	*BNIP3*	Yes	Autophagy and metabolic reprogramming	Induces autophagy and sensitizes carcinoma cells to chemotherapy	205
Gal-3	ALG-2	No	Autophagy and metabolic reprogramming	Prevents lysosomal leakage by stabilizing membranes	206
	Rag GTPase	Yes	Autophagy and metabolic reprogramming	Enhance glycolytic gene expression in tumor cells	212
Gal-8	NDP52, GALTOR complex	No	Autophagy and metabolic reprogramming	Xenophagy, maintains homeostasis, autophagy initiation via GALTOR	207–209
Gal-9	USP9X	No	Autophagy and metabolic reprogramming	Promotes autophagy during lysosomal stress	210,211
Fibrosis and ECM remodeling					
Gal-1	*TGF-β1/Smad*	Yes	Pancreatic fibrosis	ECM synthesis, migration, myofibroblast proliferation	213,214
	FAK1	Yes	Lung injury models	Collagen deposition, epithelial remodeling	215
Gal-3	*TGF-β receptor II /Smad*, β-catenin	Yes	Pulmonary fibrosis	Increases catenin nuclear translocation, induces fibrosis	217
	α-SMA induction	Yes	Renal fibrosis	Promotes fibroblast activation and differentiation	219
	IKKβ/NF-κB, iNOS	Yes	Cardiac fibrosis	ECM accumulation, cardiac fibrosis – positive feedback loop, M2 macrophage polarization	220

Table 1. continued

Galectin	Interacting partners*	Promote disease yes/no	Biological process/condition	Effect	Ref.
Cancer stemness and differentiation					
Gal-1	STAT3/HOXA5	Yes	Glioblastoma	GSC maintenance, treatment resistance	225
Gal-3	*EGFR/c-Myc/Sox2, Notch1*, integrin αvβ3/KRAS	Yes	Breast cancer, lung cancer, lung cancer, pancreatic cancer	CSC stemness, differentiation, treatment resistance, tumor progression	229,230,232
Gal-7	JNK1	Yes	Keratinocyte differentiation	Epithelial maturation to regulate differentiation	80
Gal-9	*TGF-β3/Smad2*	Yes	Chondrogenesis in mesenchymal stem cells	Induces chondrogenesis to drive cartilage lineage commitment	234
Gal-12	*C/EBPβ, C/EBPα, PPARγ*	Yes	Lipid droplets – adipocyte differentiation	Adipocyte differentiation, promotes expression of adipogenic transcription factors	235
		No	APL	Suppress neutrophilic differentiation	236
Therapy resistance and drug response					
Gal-1	*H-Ras/Raf-1/ERK, HIF-1α, VEGF*	Yes	Driven by hypoxia, radiation inducing $CD8^+$ T cell	Radiotherapy resistance, angiogenesis, apoptosis, immune suppression	237,238
	VEGFR2	Yes	Tumors resistant to VEGF inhibition, sustaining angiogenesis	Anti-VEGF- resistance	167
	BRAF	Yes	Melanoma	Compensatory immunosuppressive mechanism by inducing apoptosis of interacting T cells	239
	HOXA5	Yes	Glioblastoma	Ionizing radiation resistance	225
	Integrin β1, *FAK/c-Src/ERK/STAT3*	Yes	TNBC	Doxorubicin resistance, anti-apoptotic survivin expression	240,241
	Wnt/ β-catenin, MDR1	Yes	Esophageal squamous cell carcinoma	Paclitaxel resistance	242
Gal-3	*PI3K/AKT, Notch1*	Yes	HER2-breast cancer	Trastuzumab resistance	244
Gal-9	TIM-3, *calpain-caspase 1, STING-IFN-I*	Yes	Immunosuppressive tumor microenvironment	Calcium influx, T cell apoptosis, deactivate immune checkpoint inhibitors, suppress DC activation	163,204,245
Genome reprogramming					
Gal-1	*STAT3*/HOXA5	Yes	Glioblastoma	Reprogram GSC transcriptional landscape	225
Gal-3	*Wnt/β-catenin, PI3K/Akt, Wnt/STAT3*, STAT3 SH2 domain	Yes	Gastric cancer	Enhance STAT3 phosphorylation and transcriptional output	246,247
Gal-9	NF-IL6 (C/EBPβ)	No	Inflammatory conditions	Increase transcription of immune response genes	248

HNSCC head and neck squamous cell carcinoma, *MM* metastatic malignant melanoma, *TNBC* triple-negative breast cancer, *AML* acute myeloid leukemia, *CSC* cancer stem cell, *EMT* epithelial-to-mesenchymal transition, *MMP* matrix metalloproteinases, *ECM* extracellular matrix, *GSC* glioblastoma stem cell, *APL* acute promyelocytic leukemia, *DC* dendritic cell, *BTSC* brain tumor stem cell
*Note: italicized interacting partners indicate functional and non-physical binding partners

and activating FAK, which transactivates epidermal growth factor (EGFR) signaling and engages proteasome-dependent pathways that stabilize EMT transcription factors such as Snail and β-catenin. These events induce partial EMT, MMP expression, and migratory phenotypes in epithelial cells.[184]

Although less extensively studied, Gal-9 can influence cell adhesion and motility through interaction with CD44 and hyaluronan. Katoh et al. showed Gal-9 inhibits the interaction between CD44 and hyaluronan, reducing cell adhesion and migration in a murine model of allergic asthma.[185] Similar mechanisms may operate in cancer, where Gal-9 limits tumor cell adhesion to vascular endothelium and extracellular matrices, thereby suppressing metastatic spread[186] (Table 1).

Overall, galectins, particularly Gal-1, Gal-3, and Gal-8, promote tumor invasion and metastasis by regulating integrin signaling, cytoskeletal dynamics, matrix remodeling, and EMT programs.

Apoptosis and cell survival

Galectins exhibit context-dependent effects on cell fate, functioning as either pro-apoptotic or pro-survival factors. In tumor cells, Gal-1 generally acts as a pro-survival molecule, and its inhibition triggers apoptosis.[187,188] In contrast, in immune cells, Gal-1 functions predominantly to promote apoptosis by binding glycoproteins such as CD45, CD43, and CD7 on activated T cells.[156,189–192] Gal-1 also activates AP-1 (c-Jun) transcription factor, further amplifying pro-apoptotic signaling.[193]

Gal-3 contains a conserved NWGR motif homologous to the BH1 domain of Bcl-2 family proteins and functions primarily as an anti-apoptotic regulator.[194,195] It regulates the balance between pro-apoptotic and anti-apoptotic proteins,[194,196] inhibits mitochondrial outer membrane permeabilization,[196,197], and suppresses Cytochrome C release,[197] thereby, blocking intrinsic apoptosis.[194–198] However, extracellular Gal-3 can induce apoptosis in T cells through interaction with cell-surface glycoproteins, including CD45 and CD71.[199]

Gal-4 also regulates apoptosis in a context-dependent manner. In T cells, Gal-4 binds the CD3 complex via carbohydrate-dependent interactions, inducing apoptosis through a calpain-dependent but caspase-independent pathway.[200] In colorectal cancer cells, Gal-4 downregulates the anti-apoptotic protein Bcl-2, promotes cell cycle arrest, and enhances camptothecin-induced apoptosis while suppressing Wnt-β-catenin signaling.[201] In pancreatic cancer, extracellular Gal-4 induces T cell apoptosis by binding N-glycosylated CD3ε/δ, promoting immune evasion and tumor progression.[200]

Gal-7 is a direct transcriptional target of p53 and contributes to DNA-damage-induced apoptosis. Acting upstream of JNK1, it sensitizes cells to UV- or genotoxic stress by promoting mitochondrial Cytochrome C release and caspase-3 activation.[202]

Gal-8 induces apoptosis in T lymphocytes through phospholipase D (PLD)-phosphatidic acid (PA) signaling, which increases ERK1/2 phosphorylation, suppresses protein kinase A activity (PKA), and promotes Fas ligand expression, leading to caspase-dependent apoptosis.[203]

Gal-9 exhibits complex and context-dependent functions. In effector T cells, Gal-9, interacting with TIM-3, induces calcium influx and activation of the calpain–caspase-1 pathway, promoting immune cell death and tolerance.[163,204] In contrast, in leukemic stem cells, Gal-9 can promote survival in a TIM-3 dependent mechanism[98] (Table 1).

Overall, galectins regulate apoptosis through diverse mechanisms. Several members, including Gal-1, Gal-3, and Gal-9, exhibit context-dependent duality, promoting apoptosis in immune cells while suppressing tumor cell survival, thereby contributing to tumor progression.

Autophagy and metabolic reprogramming

Several galectins have emerged as key regulators of cellular homeostasis by modulating autophagy and metabolic signaling pathways. Gal-1 inhibits PI3K/AKT/mTOR signaling, inducing BNIP3-dependent autophagy and sensitizing hepatocellular carcinoma cells to chemotherapeutic agents.[205] Gal-3 contributes to lysosomal membrane repair by interacting with apoptosis linked gene-2 (ALG-2) and components of the endosomal sorting complex required for transport (ESCRT) machinery, including ALG-2-interacting protein X (ALIX) and charged multivesicular body protein 4B (CHMP4B), thereby stabilizing damaged lysosomal membranes and preventing lysosomal leakage.[206]

Gal-8 functions as a cytosolic danger receptor that detects damaged endosomes or lysosomes and recruits autophagy adapters such as nuclear dot protein 52 (NDP52), initiating selective autophagy (xenophagy), to facilitate the clearance of intracellular pathogens and damaged organelles.[207,208] Gal-8 also regulates metabolic signaling through the AMPK/mTOR pathway. Upon lysosomal damage, Gal-8 binds exposed glycans on ruptured lysosomes and recruits the GALTOR complex, a lysosomal mTOR regulatory assembly composed of SLC38A9, Ragulator, and Rag GTPases. This interaction disrupts Ragulator-Rag GTPases signaling, leading to mTORC1 inactivation and induction of autophagy.[209,210]

Gal-9 also promotes autophagy during lysosomal stress by activating AMPK. It displaces ubiquitin-specific peptidase 9 X-linked (USP9X) from the transforming growth factor-beta-activated kinase 1 (TAK1) complex, enabling K63-linked ubiquitination of TAK1, which activates AMPK, suppresses mTORC1, and enhances autophagic flux.[210,211] In contrast, Gal-3 can promote mTORC1 activation under inflammatory stimuli such as lipopolysaccharide (LPS) by stabilizing Rag GTPase–Ragulator complexes, and increasing the expression of glycolytic gene including GLUT1, HK2, and PKM, in tumor cells.[212]

Through these mechanisms, galectin-driven regulation of autophagy and metabolic reprogramming enable tumor cells to adapt to hypoxia, nutrient deprivation, and oxidative stress, thereby supporting survival, sustained proliferation, and therapy resistance[205] (Table 1).

Fibrosis and ECM remodeling

Fibrogenic responses in cancer and chronic disease are partly driven by galectin signaling, particularly by Gal-1, Gal-3, and Gal-8. In pancreatic stellate cells and pancreatic ductal adenocarcinoma (PDAC) stroma, Gal-1 expression is upregulated and promotes fibrosis through TGF-β1/Smad-dependent signaling, enhancing extracellular matrix (ECM) synthesis, migration, and myofibroblast proliferation.[213,214] In lung injury models, hypoxia-induced Gal-1 activates FAK1 phosphorylation, promoting collagen deposition and epithelial remodeling, effects reversed by Gal-1 inhibition.[215] Gal-3 is a major regulator of fibrosis and ECM remodeling through both canonical TGF-β/Smad signaling and Smad-independent pathways. TGF-β1 signaling activates Smad2/3-Smad4 complexes that induce transcription of profibrotic genes such as *ACTA2* (α-smooth muscle actin (α-SMA)), *COL1A1*, and fibronectin, leading to myofibroblast differentiation and ECM deposition.[216,217]

Gal-3 promotes fibrosis in both liver and lung by facilitating myofibroblast activation and extracellular matrix production without directly altering Smad2/3 phosphorylation.[217,218] In lung fibrosis, it further potentiates TGF-β-induced β-catenin activation and nuclear translocation by inhibiting GSK-3β phosphorylation, thereby promoting fibrogenesis. Gal-3 also activates β-catenin signaling independently of Smad pathways, contributing to EMT and fibroblast activation. In experimental pulmonary fibrosis, Gal-3 inhibition reduces β-catenin nuclear translocation and attenuates fibrotic remodeling.[217]

Gal-3 can drive fibrosis in other organs. In renal fibrosis, Gal-3 secreted by macrophages acts in a paracrine fashion to promote fibroblast activation and myofibroblast differentiation (via α-SMA induction), independently of TGF-β/Smad pathways.[219] In cardiac fibrosis, Gal-3 enhances collagen synthesis and cross-linking.

Myocardial infarction stimulates Gal-3 secretion from inflammatory cells, which drives myofibroblast activation via the TGF-β pathway. Furthermore, Gal-3 can promote M2 macrophage polarization and arginase-1 expression, establishing a positive feedback loop that amplifies fibrotic remodeling. In contrast, activation of the IKKβ/NF-κB signaling pathway increases iNOS expression, associated with M1 macrophages.[220] Genetic or pharmacological inhibition of Gal-3 reduces myofibroblast activation and fibrogenesis.[221]

In contrast to Gal-1 and Gal-3, Gal-8 appears to play a protective role in fibrotic remodeling. In models of acute kidney injury (AKI), *LGALS8* knockout mice exhibit exacerbated renal fibrosis, with elevated collagen I/III deposition and increased Th17 immune infiltration, suggesting that Gal-8 normally limits maladaptive wound repair and fibrosis progression[222] (Table 1).

Overall, galectins exert distinct and sometimes opposing roles in fibrosis. Gal-1 and Gal-3 function predominantly as fibrotic mediators that promote fibroblast activation, myofibroblast differentiation, and ECM deposition across multiple organs, whereas Gal-8 restrains excessive fibrosis and tissue remodeling. These findings underscore the complexity of galectin signaling and the importance of defining galectin-specific functions when developing targeted anti-fibrotic therapies.

Cancer stemness and differentiation

Cancer stem cells (CSCs) represent a subpopulation of tumor cells with the ability to self-renew, initiate tumors, and drive metastasis and therapeutic resistance.[223] Increasing evidence indicates that galectins are important regulators of CSC maintenance and plasticity by integrating glycan-mediated extracellular interactions with intracellular signaling pathways that govern stemness programs.[224,225]

LGALS1, the gene encoding Gal-1, is upregulated in glioblastoma and CSC population, and its elevated expression correlates significantly with poor patient prognosis.[225] *LGALS1* is a direct transcriptional target of signal transducer and activator of transcription 3 (STAT3), a key transcription factor required for EMT, tumor malignancy, and aggressiveness. Importantly, the encoded protein, Gal-1, forms a complex with homeobox A5 (HOXA5), another transcription factor, whose expression correlates with the gain of chromosome 7 and a more aggressive phenotype of glioma.[226] Genetic deletion of *LGALS1* or pharmacological blockage of Gal-1, impairs glioblastoma stem cell (GSC) and sensitizes the tumors to IR therapy. Notably, this study also illustrated that Gal-1 signaling supports stemness and glioblastoma growth via reprogramming transcriptional networks that drive the most aggressive mesenchymal glioblastoma.[225]

Gal-3 is one of the most extensively studied galectins in CSC biology and functions as a central regulator of stemness across multiple tumor types. Proteomic analyses of tumor spheres derived from MCF-7 breast cancer cells, commonly used as an in vitro CSC model, revealed consistent upregulation of Gal-3 alongside CSC markers. Pharmacological inhibition of Gal-3 using *N*-acetyllactosamine (LacNAc) significantly reduced tumor-sphere formation, implicating Gal-3 as essential for CSC self-renewal.[224] Mechanistically, Gal-3 promotes stemness via stabilization of β-catenin, preventing its proteasomal degradation and sustaining Wnt/β-catenin signaling, a critical pathway,[227] regulating the balance between CSC self-renewal and differentiation.[228] In lung cancer, Gal-3 further promotes CSC phenotypes by activating the EGFR/c-Myc/Sox2 signaling axis. Sox2 is a key pluripotency factor required for stem cell maintenance, and its upregulation reinforces the CSC transcriptional program. Notably, the stemness factor Oct4 upregulates Gal-3 expression, establishing a positive feedback loop that sustains the CSC phenotype.[229] Gal-3 also facilitates Notch1 activation by promoting cleavage of the Notch1 intracellular domain (NICD), thereby sustaining Notch signaling, which is essential for CSC identity and asymmetric cell division.[230]

Regulation of Gal-3 expression within CSCs is tightly controlled by transcriptional and post-transcriptional mechanisms. In GSCs, Gal-3 expression is upregulated by Krüppel-like factor 4 (KLF4), a pluripotency-associated transcription factor. Conversely, microRNA-152 (miR-152) acts as a negative regulator of Gal-3 by targeting its promoter and suppressing downstream MEK1/2 and PI3K signaling.[231] Through these mechanisms, Gal-3 integrates multiple oncogenic signaling networks that collectively sustain stemness and tumor aggressiveness. In addition, Gal-3 cooperates with integrin αvβ3 and oncogenic KRAS to activate NF-κB signaling, further promoting CSC maintenance, and tumor aggressiveness in lung and pancreatic tumors.[232]

Beyond maintaining CSC populations, galectins regulate cell differentiation programs in both normal and malignant contexts. Interestingly, Gal-3 can promote lineage differentiation under certain conditions. For example, Gal-3 derived from human umbilical cord mesenchymal stem cell (hucMSC) exosomes promotes fibroblast-to-myofibroblast differentiation. Mechanistically, Gal-3 increases β-catenin accumulation and nuclear translocation, leading to enhanced expression of myofibroblast markers such as α-SMA, collagen I, and periostin.[233]

Other galectins primarily function as regulators of lineage commitment. Gal-7 drives keratinocyte differentiation via activation of the JNK1–miR-203–p63 pathway, a pathway critical for epidermal differentiation and epithelial homeostasis.[80] Gal-9 enhances TGF-β3-induced chondrogenesis in mesenchymal stem cells by synergistically promoting Smad2 phosphorylation and facilitating cartilage lineage commitment.[234] Gal-12 also plays an important role in differentiation, particularly in adipogenesis. Knockdown of Gal-12 impairs the induction of critical adipogenic transcription factors (C/EBPβ, C/EBPα, and PPARγ) and adipocyte differentiation markers (such as aP2 and adipsin).[235] However, Gal-12 can exert context-dependent effects; in hematopoietic malignancies, Gal-12 suppresses neutrophilic differentiation in acute promyelocytic leukemia (APL) cells; its knockdown enhances myeloid maturation,[236] suggesting that Gal-12 maintains an undifferentiated state in select contexts (Table 1).

Collectively, these findings highlight the multifaceted roles of galectins in regulating both cancer stemness and cellular differentiation. Gal-1 and Gal-3 primarily control CSC maintenance across multiple tumor types by activating stemness-associated signaling networks such as STAT3, Wnt/β-catenin, EGFR, Notch, and NF-κB, thereby promoting tumor initiation, metastasis, and therapeutic resistance. In contrast, other galectins, including Gal-7, Gal-9, and Gal-12, are involved in controlling lineage commitment and differentiation processes, although Gal-12 can also contribute to an undifferentiated state in certain malignancies. These context-dependent roles underscore the importance of galectin-mediated signaling in tumor cell plasticity and highlight galectins as promising therapeutic targets for disrupting CSC maintenance and overcoming treatment resistance (Table 1).

Therapy resistance and drug response

Galectins have been directly implicated in mediating resistance to various cancer therapies through distinct signaling mechanisms. Among them, Gal-1 contributes to therapy resistance across several cancer types. It promotes radiotherapy resistance by enhancing H-Ras/Raf-1/ERK pathway activation, facilitating DNA damage repair, and suppressing apoptosis.[237] Tumor hypoxia and radiation upregulate Gal-1 expression, leading to apoptosis of $CD8^+$ T cells and subsequent immune suppression, which impairs radiotherapy efficacy.[238] Gal-1 has been linked to resistance to anti-VEGF therapy. In tumors resistant to VEGF inhibition, Gal-1 is upregulated and binds glycosylated VEGFR2 on endothelial cells, activating VEGF-like signaling that sustains angiogenesis. Disruption of Gal-1-VEGFR2 interaction restores sensitivity to anti-VEGF therapy.[167] Furthermore, BRAF inhibition increases Gal-1 expression in melanoma cells, and elevated Gal-1 plasma levels have been observed in patients progressing on BRAF/MEK inhibitors. This upregulation likely represents a compensatory immunosuppressive mechanism, as Gal-1 induces apoptosis of interacting T cells, and promotes immune escape.[239] Additionally, Gal-1 promotes immune evasion by excluding T cells from the tumor microenvironment, thereby reducing response to immunotherapy.[158] In glioblastoma, Gal-1 supports GSC survival and contributes to resistance to ionizing radiation through interaction with the transcription factor HOXA5.[225] In TNBC, extracellular Gal-1 promotes doxorubicin resistance by interacting with integrin β1, and activating downstream FAK/c-Src/ERK/STAT3 signaling, ultimately upregulating the anti-apoptotic protein survivin. Disruption of this signaling axis via Gal-1 knockdown restores drug sensitivity.[240,241] Similarly, in esophageal squamous cell carcinoma, Gal-1 enhances Wnt/β-catenin signaling and increases MDR1 expression, promoting paclitaxel resistance, which can be reversed with the Gal-1 inhibitor OTX008.[242] Gal-3 has also been implicated in therapy resistance across multiple cancers. In pancreatic adenocarcinoma, Gal-3 promotes gemcitabine resistance via tumor-stromal interactions and suppression of mitochondrial oxidative phosphorylation, while its inhibition restores chemosensitivity.[243] In HER2-positive breast cancer, elevated Gal-3 expression contributes to trastuzumab resistance by activating the PI3K/AKT pathway and enhancing CSC-like properties through Notch1 signaling. Conversely, Gal-3 depletion improves therapeutic response.[244]

Gal-9 primarily mediates therapy resistance through immune regulatory mechanisms. By binding to the TIM-3 receptor on T cells, Gal-9 induces calcium influx and activates the calpain–caspase-1 pathway,[163,204] leading to T cell exhaustion and apoptosis. This crosstalk weakens anti-tumor immunity and reduces the effectiveness of immune checkpoint inhibitors.[163,204] Additionally, tumor-intrinsic STING-IFN-I signaling can drive Gal-9-mediated resistance to EGFR tyrosine kinase inhibitors by suppressing DC activation and $CD8^+$ T cell infiltration, thereby fostering an immunosuppressive tumor microenvironment.[245] Although evidence linking other galectins to therapy resistance remains limited, Gal-1, Gal-3, and Gal-9 contribute to therapy resistance across multiple cancer types. Targeting galectin-mediated mechanisms, therefore, represents a promising strategy to enhance treatment efficacy and overcome drug resistance.

Genome reprogramming

Galectins have emerged as upstream modulators of oncogenic transcriptional programs, functioning not only as extracellular GBPs but also as intracellular regulators of transcription factor activity and chromatin-dependent gene expression. Increasing evidence suggests that galectins can influence transcriptional networks by forming complexes with transcription factors or by modulating signaling pathways that control transcriptional activation. For example, in glioblastoma, we demonstrated that Gal-1 forms a complex with HOXA5, enhancing its DNA-binding activity and reprogramming the transcriptional landscape of GSCs.[225] This interaction promotes transcriptional programs associated with stemness, survival, and therapy resistance, suggesting a model whereby Gal-1 may function as a transcriptional co-regulator that links extracellular stress signals to stem cell–associated gene expression programs.

Gal-3 has similarly been shown to regulate transcription indirectly. In several cancer models, Gal-3 enhances Wnt/β-catenin signaling, by promoting nuclear accumulation of β-catenin and upregulating canonical target genes such as cyclin D1 and c-Myc. This effect is mediated in part through PI3K/Akt-dependent inhibition of GSK-3β.[246] It also facilitates Wnt/STAT3 crosstalk by directly binding to the STAT3 SH2 domain, enhancing STAT3 phosphorylation and transcriptional output in gastric cancer.[247] Through these mechanisms, Gal-3 may amplify oncogenic transcriptional circuits that coordinate proliferation, survival, and stem-like phenotypes.

Gal-9 provides another example of galectin-mediated transcriptional regulation through direct interaction with transcription factors. Gal-9 associates with NF-IL6 (C/EBPβ) in monocytes and promotes transactivation of pro-inflammatory cytokine genes such as IL-1α and IL-1β. This activity is enhanced by LPS stimulation and involves nuclear translocation of Gal-9, supporting the concept that Gal-9 may function as a transcriptional co-activator within inflammatory gene regulatory networks.[248] Given the central role of inflammatory signaling in tumor progression and immune modulation, such interactions may contribute to tumor-associated inflammation and remodeling of the tumor microenvironment.

Beyond direct transcriptional regulation, galectin expression itself is tightly controlled by epigenetic mechanisms, creating feedback loops between glycan-mediated signaling and chromatin state. In mesenchymal-like glioblastoma, histone deacetylase HDAC7 deacetylates H3K27 at the SOX8 promoter, repressing SOX8 expression. Loss of SOX8 permits JUN-dependent transcriptional activation of *LGALS3*, leading to increased Gal-3 expression and activation of integrin ITGB1 signaling, essential for tumor cell invasion. This HDAC7-H3K27ac-SOX8-JUN-*LGALS3*-ITGB1 regulatory axis highlights how histone modifications at galectin loci control downstream adhesion and signaling pathways.[249]

DNA methylation plays an important role in regulating galectin expression. For instance, *LGALS1* promoter hypomethylation correlates with elevated Gal-1 expression in amyotrophic lateral sclerosis (ALS),[250] suggesting that epigenetic derepression may contribute to pathological upregulation of Gal-1 in disease contexts. Conversely, hypermethylation at CpG sites within *LGALS3* decreases Gal-3 expression in glioblastoma and is associated with improved overall survival in patients with the proneural subtype of glioblastoma.[251] Similarly, epigenetic regulation of *LGALS9* through reduced histone H3K9 and H3K14 acetylation at the promoter has been observed in cervical cancer cells, associated with decreased Gal-9 expression.[252] These observations suggest that histone acetylation and DNA methylation at galectin loci serve as key regulatory nodes linking chromatin remodeling to glycan-binding protein expression.

Collectively, these findings position galectins as both regulators and targets of transcriptional and epigenetic networks. By modulating transcription factor activity, intracellular signaling cascades, and chromatin-dependent gene regulation, galectins may integrate extracellular glycan signals with intracellular transcriptional responses. It is tempting to speculate that galectins function as molecular bridges between glycosylation-dependent cell surface signaling and nuclear transcriptional programs, thereby enabling tumors to rapidly adapt their gene expression landscape in response to microenvironmental cues such as hypoxia, inflammation, and therapy-induced stress. Future studies will be required to determine whether galectins more broadly participate in chromatin remodeling complexes or influence higher-order genome organization, which could further explain their ability to coordinate transcriptional plasticity and tumor evolution.

ROLE OF GBPS IN DISEASE PATHOGENESIS

Cancer

According to global cancer statistics, cancer is a leading cause of death worldwide, accounting for approximately 10 million deaths in 2020. Among these, a substantial proportion are attributed to lung, colorectal, liver, stomach, and breast cancers.[253] As discussed in section 4.3, galectins are heavily involved in regulating oncogenic signal transductions, and their inhibition is associated with improved patient outcomes in several malignancies. Among the family members, Gal-1 and -3 have been the most extensively studied for their ability to activate oncogenic signaling. Other galectins, however, display context-dependent functions, acting either as pro- or anti-tumor modulators depending on factors such as cancer type, subcellular localization, glycosylation status of interacting receptors, and the composition of the tumor microenvironment.

Pro-tumorigenic galectins, particularly Gal-1, and Gal-3, play central roles in tumor growth, invasion, and metastasis. The broad presence of Gal-1 across many cancer types and its strong association with tumor progression are largely attributed to its ability to interact with multiple cell-surface receptors and signaling molecules.[47,50] In gliomas, Gal-1 interacts with activated H-Ras-GTP, integrin β1, and VEGFR2, thereby activating the PI3K/AKT that promote tumor cell proliferation and survival.[50,254–256] Similarly, Gal-1 has been linked to MAPK/ERK signaling pathways and to PI3K/AKT in prostate cancer.[257,258] In TNBC, Gal-1 binds integrin β1 to activate the FAK/c-Src/ERK/STAT3 signaling cascade, enhancing tumor cell survival and invasive potential.[259] Notably, Gal-1 expression is strongly induced under hypoxic conditions, and can act as a transcriptional downstream effector protein of HIF-1α in colon cancer and renal cell carcinoma, where it promotes AKT and later mTOR via phosphorylation.[258] In pancreatic ductal adenocarcinoma, Gal-1 shapes the tumor microenvironment by directing pancreatic stellate cells to secrete IL-6, which in turn activates the IL-6/JAK/STAT3 pathway to increase VEGF expression and angiogenesis. Concurrently, Gal-1 enhances NRP-1 expression, promoting immunosuppression and sustaining its hypoxic microenvironment that facilitates tumor invasion and metastasis.[214,260] These findings suggest that Gal-1 may function as a central signaling hub that integrates hypoxia, stromal activation, and oncogenic receptor signaling within the tumor microenvironment.

Similarly, Gal-3 is broadly expressed in various cancers and primarily activates oncogenic signaling through its ability to bind glycosylated receptors on the cell surface. Gal-3 interacts with receptor tyrosine kinases (RTKs), including EGFR and VEGFR, stabilizing receptor clustering and prolonging downstream signaling.[261,262] In epithelial cancer cells, Gal-3 binds MUC1, which enhances EGFR dimerization, and sustains EGFR signaling.[261] Through these interactions, Gal-3 promotes cancer stemness in lung cancer via the EGFR/c-Myc/Sox2 pathway.[229,263] In addition, Gal-3 modulates Notch1 in breast cancer,[244] PI3K/AKT/GSK-3β/β-catenin in HCC,[264] and Wnt/β-catenin, K-Ras/Raf/ERK1/2 in colon cancer,[265] collectively facilitating malignancy, angiogenesis, and metastasis. Given its ability to crosslink glycosylated receptors and scaffold signaling complexes, Gal-3 may act as an extracellular organizer of oncogenic receptor networks, amplifying signaling outputs that drive tumor progression.

In contrast, several galectin family members exhibit dual roles or context-dependent roles in tumor biology. For instance, Gal-4 functions as a tumor suppressor in normal colon tissue by inhibiting Wnt/β-catenin and IL-6/NF-κB/STAT3 signaling.[71,201] However, in prostate cancer, Gal-4 interacts with EGFR, HER2, and HER3 to activate AKT and fibronectin pathways that promote EMT, metastasis, and invasion.[263,266,267] Gal-7 also demonstrates dual functionality. Gal-7 promotes tumor progression in cervical adenocarcinoma and hypopharyngeal/laryngeal squamous cell carcinoma by increasing MMP-9 expression via the MAPK signaling pathway.[268] Yet, ectopic expression of Gal-7 in colorectal cancer cells suppresses tumor growth.[269] Gal-8 regulates cytokine and chemokine expression in breast, prostate, and lung cancers through integrin-mediated FAK/AKT/ERK/JNK/NF-κB signaling,[87,89] yet in other contexts it can inhibit cell adhesion, suppress migration, or induce apoptosis.[270,271] These contrasting observations highlight how galectin activity is highly dependent on cellular context, receptor glycosylation patterns, and microenvironmental conditions.

Gal-9 also displays complex roles in cancer progression. Gal-9 facilitates AML progression through PKC/mTOR-mediated TIM-3 release.[91,98] Conversely, in solid tumors, Gal-9 can enhance anti-tumor immunity by promoting CD8⁺ T cell-mediated cytotoxic responses.[272] Moreover, Gal-9 can directly induce apoptosis in melanoma and other cancer cells through mitochondrial pathways.[273,274] Similarly, Gal-12 exhibits dual functions, promoting survival in AML when hypomethylated,[275] yet inducing apoptosis and metabolic changes in adipocytes and cervical cancer via AKT, ERK, and NF-κB activation, while simultaneously suppressing STAT3.[117]

Although the roles of several galectins are increasingly appreciated, other family members remain relatively understudied in cancer. Gal-2 modulates cancer progression through interactions with GM1 in neuroblastoma cells and MUC1 in epithelial cells.[55] Gal-10 has been implicated in eosinophilic leukemia.[104] Gal-13 increases PD-L1, TNF-α, and ROS in neutrophils, potentially aiding immune evasion.[276] Computational analyses revealed that Gal-16 is expressed in breast cancer, chronic myeloid leukemia, and B cell lymphoma,[127] though its molecular mechanisms remain unclear.

Overall, galectins exert both pro- and anti-tumorigenic effects depending on cancer type, localization, glycosylation context, and interacting partners. Their ability to regulate key oncogenic pathways such as PI3K/AKT, MAPK/ERK, JAK/STAT3, and Wnt/β-catenin underscores their potential as both biomarkers and therapeutic targets in oncology. It is therefore plausible that galectins function as glycan-sensitive signaling hubs, translating changes in cellular glycosylation and extracellular matrix composition into transcriptional and signaling outputs that promote tumor adaptation. Consequently, targeting galectin-glycan interactions may represent a promising strategy for disrupting oncogenic signaling networks and improving therapeutic outcomes in cancer.

CVDs

CVDs such as atherosclerosis, coronary artery disease, myocardial infarction (MI), ischemic stroke, and heart failure (HF), remain the leading cause of death worldwide.[65,277] Multiple galectin family members, including Gal-1, -2, -3, -4, -7, -8, and -9, participate in various processes of CVD pathogenesis,[277] and are increasingly explored as potential biomarkers and therapeutic targets.[65,278]

Gal-1 is expressed in the inflammatory infiltrate and interstitium of patients with advanced HF and cardiomyopathy.[279] Different factors such as age, metabolism, hypoxia, and inflammation, should be considered when analyzing Gal-1 expression. In Gal-1 knockout mice, mild ventricular dilation, and reduced contractility are observed, accompanied by inflammatory infiltrate composed of macrophages, lymphocytes, NK cells, and reduced Tregs, features consistent with autoimmune myocarditis.[279] These findings suggest that Gal-1 upregulation may represent a compensatory mechanism that limits immune-mediated cardiac injury. Consistent with this, Gal-1 suppresses pro-inflammatory cytokines (IL-6, TNF-α, IL-1β) and promotes macrophage polarization toward an anti-inflammatory M2 phenotype, increasing IL-10 production.[278] These effects may reduce cardiomyocyte apoptosis and support angiogenesis during tissue repair. Gal-1 may also contribute to vascular homeostasis by regulating Ca(V)1.2 calcium channel activity, thereby influencing vascular tone and blood pressure.[278,280]

Gal-2 appears to regulate arteriogenesis and inflammation. It is overexpressed in monocytes and macrophages of patients with coronary artery disease and exhibits anti-arteriogenic properties. Mechanistically, Gal-2 binds the CD14/TLR4 complex, activating the NF-κB signaling and promoting macrophage polarization towards a proinflammatory M1 phenotype, while reducing macrophage motility and apoptosis.[277] Genetic studies have also identified multiple single-nucleotide polymorphisms (SNPs) in the Gal-2 gene associated with MI susceptibility, inducing population-specific variants.[55] Notably, anti-Gal-2 treatment in ischemic heart disease mouse model, inhibits atherosclerosis and promotes arteriolar formation,[55] supporting a role for Gal-2 in vascular remodeling.

Among galectins, Gal-3 is the most extensively studied cardiovascular biomarker. Circulating Gal-3 levels strongly correlate with increased mortality in HF patients.[281] Although Gal-3 expression is low in healthy myocardium, its upregulation following cardiac injury promotes pathological remodeling and fibrosis. Secreted Gal-3 activates cardiac fibrosis, increasing expression of extracellular matrix components such as type I collagen, while inhibiting MMPs that normally regulate extracellular matrix turnover.[281] Additionally, Gal-3 regulates fibrosis by activating myofibroblasts via the TGF-β-independent and dependent pathway and may interact with syndecans to amplify profibrotic signaling in cardiac fibroblasts.[281,282]

The roles of Gal-4, -7, -8, and -9 in CVDs remains less well defined but emerging evidence suggests potential involvement. Gal-4 can bind to monocytes and stimulate macrophage secretion of TNF-α and IL-10, cytokines associated with arteriogenesis.[277] Elevated circulating Gal-4 has also been observed in HF patients with diabetes and obesity,[283] and population-based studies link higher Gal-4 levels with coronary heart disease and adverse cardiovascular outcomes.[284]

Gal-7 has been associated with MMP-9 regulation, a process implicated in both tumor progression and cardiac aging.[277,285] Beyond oncology in a similar sense, MMP-9 is also associated with cardiac aging and functional decline.[277,286,287] Although the exact mechanisms are not fully defined, these findings suggest a relevant link between Gal-7 and CVDs, suggesting a potential Gal-7 role in cardiac remodeling.

Gal-8 has a role in angiogenesis through binding to CD166 on endothelial cells, promoting capillary formation, endothelial cell migration, and angiogenesis either independently or in association with VEGF.[174,277] Finally, Gal-9 has been implicated in atherosclerosis by promoting monocyte recruitment and plaque progression.[288] It mediates inflammatory signaling in macrophages and facilitates monocyte adhesion via glycan-integrin β2 interactions on endothelial cells.[288]

Collectively, these findings indicate that galectins regulate key cardiovascular processes, including inflammation, angiogenesis, immune regulation, and fibrosis, that contribute to CVD progression. Their diverse functions and measurable circulating levels highlight their potential as diagnostic biomarkers and therapeutic targets in cardiovascular disease.

Neurodegenerative diseases (NDs)

NDs, including Alzheimer's disease (AD), ALS, Parkinson's disease (PD), and Huntington's disease (HD), are characterized by progressive loss of neuronal structure and function, driven in part by chronic neuroinflammation. Recent research has revealed that patients with NDs exhibit higher expression of galectins compared to healthy individuals.[289] In the central nervous system, Gal-1, -3, -4, -8 and -9 are expressed and regulate diverse processes including microglial activation, neuronal survival, and immune signaling. Broadly, Gal-1 and -8 are associated with neuroprotection, whereas Gal-3 and -9 promote neurodegeneration.[290]

Gal-1 plays major roles in neuronal maintenance and synaptic integrity while limiting neuroinflammation. It promotes a shift in microglial polarization, thereby suppressing neuroinflammatory mediators and supporting neuronal survival. Mechanistically, Gal-1 binds CD45 phosphatase on microglia to inhibit M1 activation.[289] In PD models, Gal-1 suppresses microglial activation through MAPK/Iκβ/NF-κB axis, leading to reduced expression of inflammatory mediators such as IL-1β, TNF-α, inducible nitric oxide synthase (iNOS), and cyclooxygenase-2 (COX-2), and improved motor function, and neuronal survival.[289,291] Blocking Gal-1's CRD is shown to abolish these protective effects.[291] Gal-1 also promotes neuronal repair following spinal cord injury via interacting with NRP-1/PlexinA4 in damaged neurons, which restores axonal regeneration and improves locomotor recovery.[292] In AD models, Gal-1 administration increased cognitive abilities, reduced microglia activation, and decreased perivascular amyloid-β (Aβ) deposition, suggesting enhanced Aβ clearance in the hippocampus.[290] Overall, these studies ultimately demonstrate the integral role of Gal-1 in neuroprotection and suggest its therapeutic potential in treatment of NDs.

Similar to Gal-1, Gal-8 exhibits neuroprotective activity through mechanisms linked to cellular stress responses and autophagy[293]. Under conditions of neuronal damage, Gal-8 detects disrupted endomembranes and recruits the autophagy adapter NDP52, initiating selective autophagy pathways that limit toxic protein aggregation.[292,293] This mechanism is particularly relevant in AD, where hyperphosphorylated tau accumulates and disrupts microtubule stability.[294] Gal-8-mediated autophagy may therefore mitigate tau aggregation and disease progression. In addition, recombinant Gal-8 increases ERK1/2 and AKT signaling in hippocampal neurons, further supporting neuronal survival pathways.[87]

Conversely, Gal-3 expression is associated with neuroinflammatory pathology and severity of NDs.[292] Activated microglia release Gal-3 following brain damage, ischemia, demyelination or encephalitis.[295] Extracellular Gal-3 amplifies microglial activation and promotes neuronal phagocytosis and Aβ aggregation. In murine models of HD, Gal-3 increases IL-1β expression, exacerbating neuroinflammation and neuronal damage.[292] Gal-3 can also activate endothelial VCAM-1, promoting monocyte infiltration and potentially contributing to neuroinflammatory diseases such as multiple sclerosis (MS).[66,296] Moreover, Gal-3 interacts with multiple immune receptors, including TLR4, MerTK, IFNGR, and TREM2, to enhance microglial reactivity and inflammatory signaling.[66,295] These findings position Gal-3 as a key driver of neuroinflammatory amplification in NDs.

Gal-9 is naturally expressed in many regions of the brain including the hippocampus, thalamus, hypothalamus, amygdala, cerebellum, cortex and olfactory bulb.[292] In secondary progressive MS, cerebral spinal fluid exhibits elevated Gal-9 levels and is positively correlated with the number of brain lesions, associated with an increase in M2 microglia and increased anti-inflammatory factors.[297] Gal-9 signaling through TIM-3 on microglia, induces pro-inflammatory responses and promotes neuronal degeneration. Consistently, blockade of TIM-3 or Gal-9 signaling in a PD mouse model, reduced microglial activation and improved motor deficits.[297] A meta-analysis further suggests that Gal-9 can promote apoptosis of encephalitogenic T cells, while increasing TNFα and IL-6 production in microglia, collectively contributing to neuroinflammation.[292] These findings suggest that Gal-9's expression is associated with neurodegeneration, although further research is required to understand Gal-9's role in pathogenesis of NDs. However, Gal-9 may also promote anti-inflammatory microglial responses[297] suggesting context-dependent functions.

Although Gal-4 is expressed in the brain, secreted in hippocampal and cortical neurons, and localized to the axonal membrane segments, research on its implication in NDs remains unclear. Gal-4 appears to be expressed in chronic lesions, where it serves to aid in remyelination and oligodendrocyte differentiation. Gal-4 expression is also upregulated in patients with PD suggesting its potential as a diagnostic marker.[298]

Collectively, these findings highlight galectins as important regulators of neuroinflammation, neuronal survival, and protein homeostasis in NDs. By modulating microglial activation, autophagy, and immune signaling pathways, galectins may act as key molecular switches linking inflammatory responses to neuronal degeneration or repair. Consequently, targeting galectin-dependent pathways represents a promising avenue for therapeutic intervention in neurodegenerative disorders.

Metabolic disorders

While galectin research has largely focused on cancer and CVDs, increasing evidence indicates that galectins also regulate metabolic disorders, including obesity, type 2 diabetes, and non-alcoholic fatty liver disease (NAFLD).[299–301] These conditions share common pathological features such as chronic low-grade inflammation, macrophage activation, and insulin resistance that are influenced by galectin-dependent signaling pathways.[299] In diabetes, persistent hyperglycemia produces advanced glycation end products (AGEs) that crosslink intracellular and extracellular proteins, contributing to complications like neuropathy, retinopathy, nephropathy and more.[299,302,303]

In obesity, pro-inflammatory macrophages accumulate within adipose tissue and alter white adipocyte metabolism, promoting insulin resistance.[301] Gal-1 expression correlates with metabolic status and adiposity. In humans, circulating Gal-1 decreases following weight loss induced by very-low-calorie diets but increases during weight gain and correlates positively with fat mass in obese children.[299,301] In diet-induced obese mice, inhibiting Gal-1 led to weight loss, reduced adipogenesis and lipogenesis, while enhancing thermogenesis and energy expenditure.[299] Gal-1 has also been linked to complications affecting the kidneys and retina.[299,301] Literature suggests that Gal-1 plays context-dependent roles in metabolic disease progression.[299,301] In longitudinal clinical studies, elevated plasma Gal-1 was a predictor of renal dysfunction and is increased in diabetic and chronic kidney disease models.[301]

In hyperglycemic conditions, the transcription factor AP4 induces Gal-1 expression, whereas pharmacological inhibition of Gal-1, suppresses AKT signaling and Gal-1 accumulation.[304] In diabetic retinopathy, Gal-1 levels are elevated in vitreous fluids, and correlate with AGEs and IL-1β. They also found that diabetes-induced retinal expression of Gal-1 can be suppressed with anti-inflammatory glucocorticoids in mice by regulating AKT, ERK1/2 and AP-1 phosphorylation.[301] Another study demonstrated that under high glucose conditions, the coincubation of renal epithelial cells and TGFβ1 (fibrotic marker), resulted in overexpression of Gal-1.[301] Experimental models further demonstrate that hypoxia and oxidative stress increase Gal-1 expression in retinal tissue, while Gal-1 silencing reduces pathological retinal neovascularization. Conversely, Gal-1 may also exert protective effects in the kidney by reducing TGF-β1-induced collagen I expression and stabilizing the calcium transporter TRPV5, thereby limiting pathological endocytosis in diabetic kidney disease.[301] Taken together, these studies show the diverse roles of Gal-1 in metabolic diseases.

Gal-3 is strongly linked to insulin resistance and T2DM. Obesity-driven insulin resistance initially triggers compensatory β-cell expansion, but prolonged metabolic stress leads to β-cell dysfunction and impaired insulin secretion.[305] Extracellular Gal-3 promotes insulin resistance by binding the insulin receptor (InsR) on hepatocytes, adipocytes, and myocytes, thereby inhibiting phosphorylation of downstream signaling molecules including IRS1, PDK1, and AKT. This reduces glucose uptake in adipose and muscle tissues while increasing hepatic glucose production.[305] Interestingly, S-glutathionylation of Gal-3, observed in both diabetic mice and humans, reduces Gal-3 binding to the InsR by approximately 75%, restoring insulin signaling and glucose uptake. Metformin treatment increases Gal-3 S-glutathionylation in diabetic mice, suggesting that modulation of Gal-3–InsR interactions may contribute to improved insulin sensitivity.[305]

Although Gal-1 and -3 are well studied, Gal-9 has also been implicated in metabolic regulation. In pancreatic β-cells, Gal-9 interacts with the glucose transporter GLUT-2, promoting glucose sensing and uptake.[306] On another note, Gal-9 was shown to induce apoptosis of TIM-3$^+$ NK cells, slowing down the development of obesity-associated NAFLD.[306]

Taken together, these findings highlight galectins as key regulators of metabolic homeostasis and inflammation. By modulating insulin signaling, immune activation, and tissue remodeling, galectins link metabolic stress to the progression of obesity-associated diseases and their complications.

Autoimmune diseases

Autoimmune diseases are characterized by the body's inability to recognize its healthy self, and attacking itself as if it was foreign, consequently leading to aberrant B cell and T cell responses, chronic inflammation, and organ dysfunction.[307] Galectins contribute to the dysregulation of immune cells and are implicated in the pathogenesis of autoimmune disorders.[308] Altered galectin expression has been reported across several conditions: systemic lupus erythematosus (SLE) exhibits elevated levels of galectins-1, -2, -3 and -9, whereas rheumatoid arthritis (RA) shows increased Gal-1, -3, -4, -7, and -9.[308] In infectious contexts such as human immunodeficiency virus (HIV) Gal-1 and -9 are shown to enhance viral infectivity.[309,310] Although the mechanisms remain incompletely understood, galectins are increasingly considered potential biomarkers and therapeutic targets in autoimmune disease.[308]

Gal-1 expression is particularly high in the lamina propria of mucosal tissues within the gastrointestinal and urogenital tracts, major sites of HIV-1 transmission.[309] Gal-1 enhances viral attachment to CD4$^+$ T cells and macrophages, facilitating infection and contributing to delayed immune responses that promote acquired immunodeficiency syndrome (AIDS).[311] In contrast, Gal-1 appears to exert immunomodulatory effects in SLE. Elevated Gal-1 expression in bone marrow–derived mesenchymal stem cells, which are used therapeutically in SLE, has been shown to restore CD4$^+$ T-cell function.[312]

Gal-3 is also elevated in SLE and has been linked to disease severity. Patients with lupus nephritis exhibit significantly higher circulating Gal-3 levels compared with those without renal involvement, and similar increases have been reported in SLE-associated dermatomyositis. Interestingly, Gal-3 expression is reduced in lesional skin from SLE patients, suggesting tissue-specific roles.[308] Additional studies on SLE patients showed that elevated Gal-9 was positively correlated with IFN signature genes, which can be used to characterize SLE. Longitudinal studies are still required to confirm Gal-9's use as a marker for IFN activity as well as its involvement in various pathways.[313]

In RA, Gal-1 is also upregulated compared to healthy individuals, although its utility as a biomarker remains unclear.[314] Synovial fluid from RA patients, shows elevated Gal-3 and its binding proteins, correlating with vascular stiffness when compared with osteoarthritis patients. Gal-3 can therefore be used as effective diagnostic tools for both RA and SLE.[308] Gal-9 is also known to regulate rheumatic inflammation by modulating apoptosis in RA. In patients with RA, suffering from joint damage and functional limitations, Gal-9 was considered to be a positive indicator.[315]

Collectively, these findings highlight the broad immunoregulatory roles of galectins across autoimmune disorders. By modulating immune cell activation, inflammatory signaling, and host-pathogen interactions, galectins may serve as both biomarkers and therapeutic targets in autoimmune disease.

THERAPEUTICALLY TARGETING GBPS

Given their diverse roles across cancer, CVDs, NDs, metabolic disorders, and autoimmune diseases, galectins have emerged as both diagnostic biomarkers and therapeutic targets. Due to their various structural CRDs, however, galectins have different glycan-binding affinities, making the development of therapies even more challenging.

Current strategies to inhibit galectin activity fall into three main categories: competitive CRD inhibitors, allosteric inhibitors, and antibody-based targeting.

The earliest galectin inhibitors were designed as glycan mimetics that competitively bind the CRD, preventing interaction with endogenous ligands.[19] These modified sugar molecules are built around sugar backbones including galactose, lactose, thiodigalactoside (TDG), talose, and lactulose, and bind to galectin's CRD to block its corresponding ligand binding[19,316–320] (Fig. 5a).

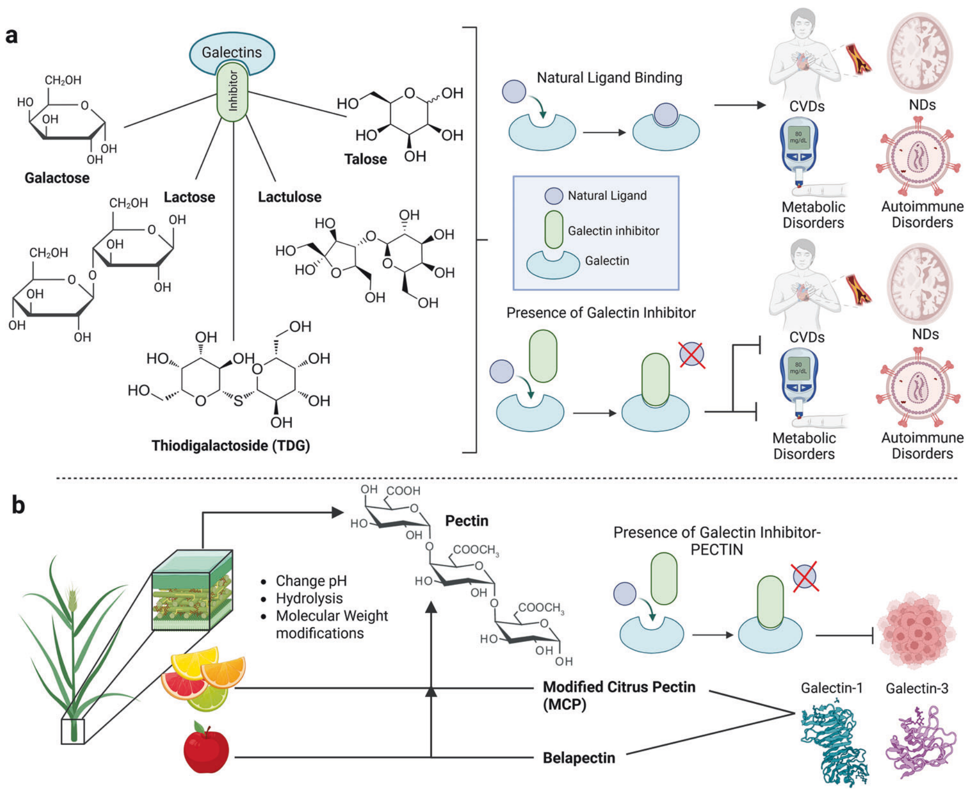

Fig. 5 Previous FDA-approved drug backbones and their mechanisms in targeting galectin downstream binding and progression of diseases. **a** Galectin inhibitors that are modified based on the galactose, lactose, TDG, lactulose, and talose backbones, preventing natural ligand binding to galectins, while allowing inhibitors to bind to galectins and prevent progression of CVDs, NDs, metabolic disorders, and autoimmune disorders. **b** Galectin binding inhibitors called pectins, derived from plant cell wall and citrus fruit. Generally indigestible to humans, they must be modified to have high affinity for Gal-1 and Gal-3 binding. Binding to galectins prevents natural ligand binding which would normally cause leukemia, multiple myeloma and other diseases. BioRender (https://www.biorender.com) was used to create this figure

To improve these molecules' inhibitory abilities, different elements like sulfur atoms were added into the glycoside linkage of TDG, to prevent breakdown by glycosidases.[19] One of the most advanced TDG derivatives is TD139 (GB0139), a high affinity Gal-3 inhibitor currently evaluated for idiopathic pulmonary fibrosis (clinical trials: NCT02257177).[321] Although selective, TD139 can also bind other galectins, including Gal-1, at lower affinity[19]. Another small molecule inhibitor includes GB1107 which suppresses lung adenocarcinoma growth and metastasis by promoting M1 macrophage polarization and $CD8^+$ T cell infiltration, thereby enhancing anti-tumor immune responses.[322,323] The related compound, GB1211, although structurally and chemically similar to GB1107, forms lasting interactions with Gal-3's CRD.[324] GB1211 has advanced into clinical evaluation and is being tested in combination with atezolizumab for non-small-cell lung cancer (NSCLC) (NCT05240131).[325] These studies highlight the growing interest in combination strategies targeting galectins alongside immune checkpoint inhibitors.

Another class of galectin inhibitors include plant-derived pectins[326] (Fig. 5b). Pectins are complex polysaccharides composed of homogalacturonan (HG), rhamnogalacturonan-I (RGI), and substituted galacturonans.[327,328] Native pectins are poorly absorbed in the human gastrointestinal tract and therefore require chemical modifications or hydrolysis to generate bioactive fragments with galectin-binding activity.[329,330]

The most extensively studied pectin is modified citrus pectin (MCP), which inhibits Gal-3 function by binding to its CRD.[328,331] A derivative of MCP, GCS-100, demonstrated activity against dexamethasone, melphalan or doxorubicin-resistant myeloma cells. Clinical trials subsequently evaluated GCS-100 in patients with chronic lymphocytic leukemia (NCT00514696) and chronic kidney disease (Phase I: NCT01717248 and Phase IIa: NCT01843790 and NCT02155673). Although the drug was generally well tolerated, only modest clinical responses were observed, and development was discontinued following regulatory and funding challenges.[332]

A related compound, Belapectin (GR-MD-02), a well-known Gal-3 inhibitor developed by Galectin Therapeutics, is a polysaccharide modified from apple pectin.[332] Although it is able to bind to Gal-1, this compound has high affinity for Gal-3.[332,333] Preclinical studies show that Belapectin enhances anti-tumor immunity, particularly when combined with anti-OX40 T cell treatment, by increasing $CD8^+$ T cell infiltration and reducing the proliferation of $FOXp3^+$ $CD4^+$ T cells, ultimately suppressing tumor progression and increasing survival in mice.[333] Recently, Belapectin has been tested in combination with Ipilimumab, a CTLA-4-blocking antibody, as well as with Pembrolizumab (Keytruda, PD-1 blocking antibody) to treat patients with melanoma and NSCLC (NCT02117362, NCT02575404, NCT04987996).[332]

Another class of inhibitors include allosteric inhibitors. Rather than directly competing for the CRD, allosteric inhibitors bind alternative sites on galectins, inducing conformational changes that disrupt ligand binding.[334,335] As previously discussed, Gal-1 is particularly difficult to therapeutically target because of its broad expression in various tissues, its roles in regulating immune and inflammatory responses and its highly conserved CRD. Synthetic peptides, therefore, have been regarded as one way of inhibiting Gal-1.[335] The synthetic peptide, Anginex, developed by *Dings* research group, is a cytokine-like peptide with anti-angiogenic activity.[336] Anginex suppresses tumor growth in a Gal-1 transgenic mouse model by disrupting Gal-1 mediated endothelial signaling.[19,335] A new and improved version of Anginex made up of a

Table 2. Summary of clinical trials using galectin inhibitors

Clinical trial (#)	Phase	Targeted galectin	Inhibitor/drugs being used	Conditions/disease	Status/results posted	Ref.
NCT05240131	I/II	Gal-3	GB1211 + Atezolizumab	NSCLC	Active / no results	325
NCT01681823	II	Gal-3	PectaSol-C	Biochemical relapsed prostate cancer	Completed / no results	359
NCT00514696	II	Gal-3	GCS-100	Chronic lymphocytic leukemia	Completed / no results	360
NCT00776802	I/II	Gal-3	GCS-100LE + etoposide/dexamethasone	Relapsed/refractory diffuse large B cell lymphoma	Withdrawn / no results	361
NCT00609817	I	Gal-3	GCS-100 + bortezomib/dexamethasone	Relapsed/refractory multiple myeloma	Terminated / no results	362
NCT00054977	I	Gal-1/ Gal-3	GM-CT-01 + /- 5-Fluorouracil	Advanced solid cancers: colorectal, lung, breast, head and neck, and prostate cancers	Completed / no results	363
NCT00388700	II	Gal-1/ Gal-3	GM-CT-01 + 5-Fluorouracil + Leucovorin + Bevacizumab	Colorectal cancer	Withdrawn/no results	364
NCT00110721	II	Gal-1/ Gal-3	GM-CT-01 + 5-Fluorouracil	Colorectal cancer	Terminated/no results	365
NCT00386516	II	Gal-1/-3	GM-CT-01 + 5-Fluorouracil	Advanced bile duct and gall bladder cancer	Withdrawn/no results	366
NCT01723813	I/II	Gal-3	GM-CT-01 + Peptide vaccination	Metastatic melanoma	Terminated/no results	367
NCT02575404	I	Gal-1/-3	GR-MD-02 + Pembrolizumab	Advanced melanoma, NSCLC, and HNSCC	Completed/no results	368
NCT04987996	II	Gal-1/-3	GR-MD-02 + Pembrolizumab + Placebo	Metastatic melanoma, HNSCC	Withdrawn/no results	369
NCT05913388	II	Gal-3	GB1211 + Pembrolizumab	Metastatic melanoma and HNSCC	Active/no results	370
NCT02117362	I	Gal-1/-3	GR-MD-02 + Ipilimumab	Metastatic melanoma	Completed/no results	371
NCT01896999	I/II	Gal-1	Brentuximab Vedotin + Nivolumab ± Ipilimumab	Relapsed/refractory Hodgkin Lymphoma	Active/no results	372
NCT01724320	I	Gal-1	OTX008	Advanced solid tumors	Unknown/no results	373
NCT04666688	I/II	Gal-9	LYT-200 + Chemotherapy (Gemcitabine/nab-paclitaxel) or Tislelizumab (Anti-PD-1)	Relapsed/refractory metastatic solid tumors	Completed/no results	374
NCT02530125	II	Gal-1	Pidilizumab	Large B Cell Lymphoma following first remission	Terminated/ results posted	375
NCT05829226	I	Gal-9	LYT-200 + Venetoclax + Azacitidine + Decitabine	AML and myelodysplastic syndrome	Completed/no results	376
NCT02257177	I/II	Gal-3	TD139	Idiopathic pulmonary fibrosis	Completed/ results posted	377
NCT03832946	II	Gal-3	GB0139	Idiopathic pulmonary fibrosis	Completed/ results posted	378
NCT04473053	I/II	Gal-3	TD139	COVID-19	Completed/ results posted	379
NCT04607655	I/II	Gal-3	GB1211	NASH and liver fibrosis	Withdrawn/no results	380
NCT05009680	I/II	Gal-3	GB1211	Hepatic impairment	Results submitted	381
NCT01960946	N/A	Gal-3	MCP/PectaSol C	Hypertension	Completed/no results	382
NCT01717248	I	Gal-3	GCS-100	Chronic kidney disease	Completed/no results	383
NCT01843790	II	Gal-3	GCS-100	Chronic kidney disease	Completed/no results	384
NCT02312050	II	Gal-3	GCS-100	Chronic kidney disease	Unknown/no results	385
NCT02155673	II	Gal-3	GCS-100	Chronic kidney disease	Completed/no results	386
NCT02333955	II	Gal-3	GCS-100	Chronic kidney disease		387

Table 2. continued

Clinical trial (#)	Phase	Targeted galectin	Inhibitor/drugs being used	Conditions/disease	Status/results posted	Ref.
					Withdrawn/no results	
NCT01899859	I	Gal-3	GR-MD-02	NASH, and advanced liver fibrosis	Completed/no results	[388]
NCT02462967	II	Gal-3	GR-MD-02	Portal hypertension + NASH, and advanced liver fibrosis	Completed/ results posted	[389]
NCT02421094	II	Gal-3	GR-MD-02	Liver fibrosis and NASH-FX	Completed/ results posted	[390]
NCT02407041	II	Gal-3	GR-MD-02	Psoriasis	Completed/ results posted	[391]
NCT04332432	I	Gal-3	GR-MD-02	Subjects with normal hepatic function and subjects with hepatic impairment	Completed/no results	[392]
NCT04365868	II/III	Gal-3	GR-MD-02	Esophageal varices in NASH cirrhosis	Terminated/ results submitted	[393]

The table explains which galectin is being targeted, if the drug was combined with others, and its treatment for cancer or other diseases, along with the status and whether or not results are available
Table adapted from Laderach, D.J., Compagno, Daniel.[19] All clinical trials were obtained from clinicaltrials.gov
NSCLC non-small-cell lung cancer, *HNSCC* head and neck squamous cell carcinoma, *NASH* non-alcoholic steatohepatitis

dibenzofuran ring and two short amino acid chains, was developed and titled 6DBF7.[335,337] This compound reduced tumor growth more efficiently than Anginex in ovarian carcinoma.[337] Additional optimization of these compounds led to development of DB16 and DB21, which proved to be significantly better at suppressing tumor growth by inhibiting Gal-1 in a noncompetitive allosteric manner.[19,50,335] One drawback of these compounds was their potential susceptibility to hydrolysis by proteases.[19] To overcome this, using the calixarene scaffold, the same group designed a non-peptidic topomimetic of Anginex, a calix[4]arene compound known as 0118/OTX008/PTX008, which allosterically binds Gal-1, thus inhibiting CRD binding. Dings et al. showed that 0118 is an effective inhibitor of tumor growth in both ovarian and melanoma xenografts. Novel calixarene-derivatives with various hydrophobic and hydrophilic characteristics are currently being developed as potential cancer therapeutic models.[50,334]

Neutralizing monoclonal antibodies (mAbs) represent another strategy to block galectin signaling. The antibody, Gal-1-mAb3, recognizes an epitope outside of Gal-1's CRD, while inhibiting LacNAc-Gal-1 interaction.[338] Similar to other inhibitors, this mAb is able to suppress angiogenesis.[338,339] Anti-Gal-1 mAb are shown to reduce the suppressive function of human and mouse $CD4^+$ $CD25^+$ Treg cells, in vitro.[340] Similarly, the Gal-3 targeting antibody inhibits cancer cells that express Mucin16 and in turn increases survival in animal tumor models.[341] The use of mAbs therefore offer promising avenues for inhibiting galectins.

Despite promising clinical trial results, significant barriers remain for clinical translation. These include the pharmacokinetic limitations, biodistribution challenges and instability of many galectin inhibitors.[19] Small-molecule inhibitors may undergo rapid renal clearance, whereas larger molecules such as antibodies primarily target extracellular galectins. Additionally, nonspecific biodistribution could inhibit galectins in healthy tissues, potentially leading to toxicity. Long-term treatment may also introduce cytotoxic or off-target effects. Another major challenge is the possibility of compensatory mechanisms among galectin family members, where inhibition of one galectin may induce functional redundancy by others. Addressing these issues will likely require the development of degradation-resistant inhibitors, improved delivery systems with tissue specificity, and rational combination therapies that account for galectin network complexity.[342,343]

CONCLUSION AND FUTURE DIRECTIONS

Extensive literature over the past several decades highlights the critical and multifaceted roles of galectins in human health and disease. These evolutionarily conserved GBPs have emerged as central regulators of cancer progression, cardiovascular disease, neurodegeneration, metabolic disorders, and autoimmune diseases. Through interactions with diverse glycoconjugates and participation in complex protein networks, galectins regulate both intracellular and extracellular signaling cascades that control fundamental cellular processes, including immune modulation, angiogenesis, metastasis, apoptosis, autophagy, fibrosis, and metabolic reprogramming. The pleiotropic functions of galectins position them as promising therapeutic targets and biomarkers across multiple diseases.[8,18]

Despite sharing structural similarities centered around conserved CRDs,[36] individual galectin family members often exhibit context-dependent and sometimes opposing roles depending on tissue type, subcellular localization, interacting partners, and disease stage. For instance, while Gal-1 and Gal-3 predominantly promote tumor progression and immune evasion in most cancer types,[47,50] galectins such as Gal-7, Gal-8, Gal-9, and Gal-12 can display either pro-tumorigenic or tumor-suppressive functions depending on the specific cellular context.[87,89,91,98,117,269–275] This functional diversity complicates therapeutic targeting strategies and necessitates understanding of galectin biology in specific disease settings. Moreover, the potential for compensatory mechanisms, where inhibition of one galectin may lead to upregulation or altered function of other family members, adds an additional layer of complexity that must be addressed in therapeutic development.[344]

Current strategies for targeting galectins remain largely preclinical, with only a limited number of inhibitors advancing to clinical evaluation. As summarized in Table 2, several galectin inhibitors including TD139/GB0139 targeting Gal-3 in idiopathic pulmonary fibrosis,[321] GB1211 targeting Gal-3 in combination with checkpoint inhibitors for cancer,[19] and GR-MD-02 (Belapectin) targeting Gal-1 and Gal-3 in liver fibrosis and various cancers[19,345] have advanced into clinical trials. However, many trials have been terminated, highlighting significant challenges in clinical translation. These challenges include suboptimal pharmacokinetics and biodistribution, lack of galectin selectivity leading to off-target

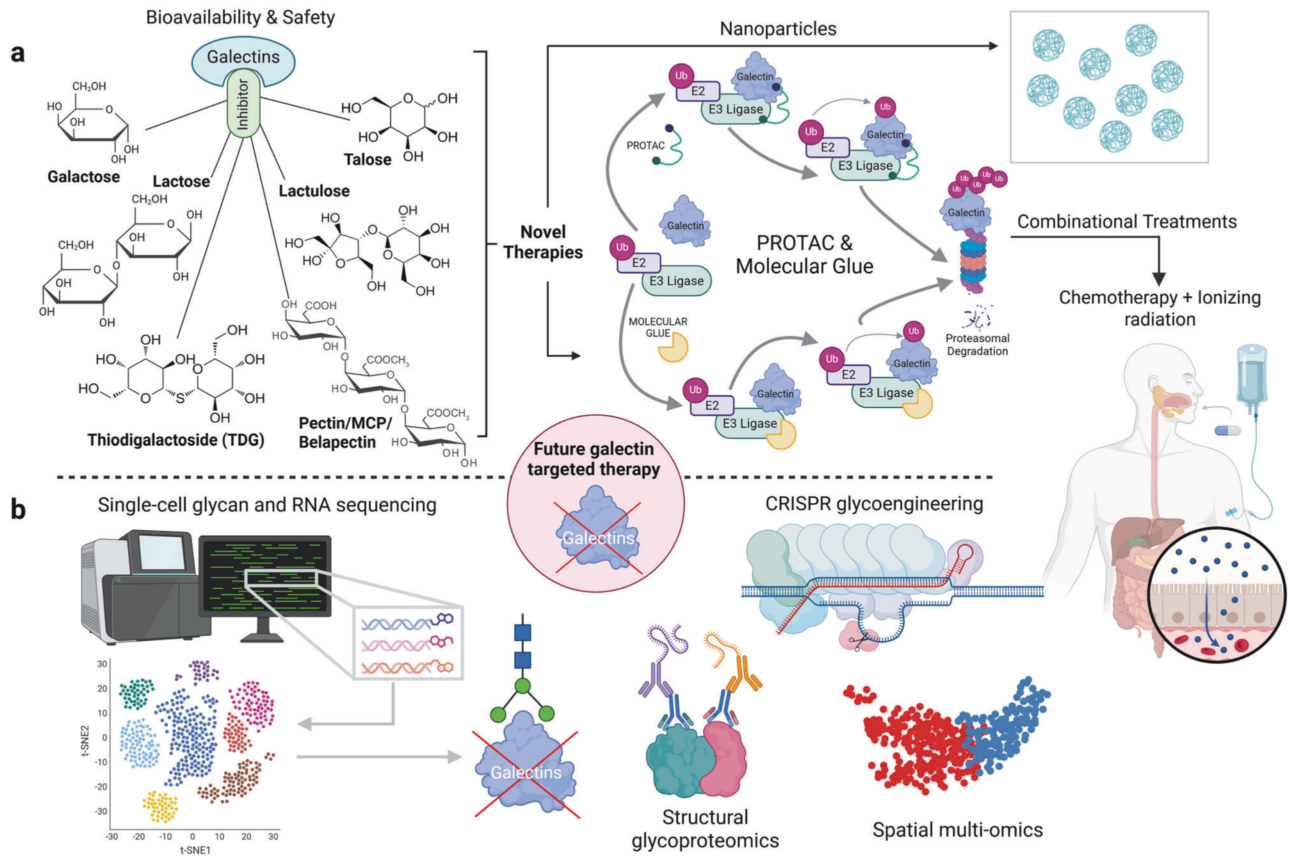

Fig. 6 Future of galectin-targeted therapy in clinical translation. **a** Studies are required to investigate the bioavailability and safety of existing galectin inhibitors. In parallel, novel selective inhibitors can be developed using nanoparticles, PROTACs, or other emerging technologies, and their efficacy can be further enhanced through combination with chemotherapy and radiotherapy. **b** Integration of novel technologies, including single-cell glycan and RNA sequencing (scGR-seq), CRISPR glycoengineering, structural glycoproteomics, and spatial multi-omics, to address cellular heterogeneity, predict galectin-glycan interactions, and achieve higher specificity while minimizing off-target effects. BioRender (https://www.biorender.com) was used to create this figure

effects, potential toxicity upon long-term administration, susceptibility to enzymatic degradation, and rapid renal clearance of small-molecule inhibitors.[19]

Future therapeutic efforts should focus on developing next-generation galectin inhibitors with improved selectivity, bioavailability, tissue-specific delivery, and safety profiles. Promising approaches include engineered nanoparticles that can encapsulate galectin inhibitors and provide targeted delivery to specific tissues or tumor microenvironments, thereby minimizing systemic toxicity. Therapeutic proteins such as recombinant galectin fragments, neutralizing antibodies, or engineered glycan-binding domains could offer enhanced specificity for particular galectin family members. Additionally, emerging targeted protein degradation technologies, including proteolysis-targeting chimeras (PROTACs) and molecular glues, optimized for specificity, bioavailability, and safety (Fig. 6a), represent revolutionary approaches that could achieve specific and sustained galectin depletion. These bifunctional molecules can recruit E3 ubiquitin ligases to galectins, marking them for proteasomal degradation.[346–348]

Beyond therapeutic development, a major challenge in the field is the incomplete understanding of galectin interaction networks and their regulation of complex cellular programs across diverse disease contexts. Galectins do not function in isolation but rather form intricate signaling hubs involving multiple glycoproteins, receptors, and downstream effectors.[36,153] Cellular heterogeneity further complicates efforts to map these interactions, as different cell populations within the same tissue may express distinct glycan patterns and galectin repertoires.[38,349] Traditional bulk analysis methods cannot capture this cell-type-specific variation, limiting the predictability of therapeutic responses and resistance mechanisms.

Single-cell glycan and glyco-RNA sequencing (scGR-seq) is a transformative technology that addresses the challenge of cellular heterogeneity by converting glycan information into a genetic readout using DNA-barcoded lectins, enabling simultaneous profiling of the glycome and transcriptome at single-cell resolution.[350] This approach reveals cell-type-specific glycan patterns and enables inference of potential galectin-glycan interactions across heterogeneous cellular populations derived from tissues or tumors.[350–353] Complementary CRISPR-based glycoengineering can generate cellular models that recapitulate defined galectin–glycan interaction patterns, providing physiologically relevant platforms for evaluating targeted therapies.[354] When integrated with structural glycoproteomics and spatial multi-omics, these technologies could provide insight into galectin function within native tissue contexts and support the rational design of galectin-targeted interventions while minimizing off-target effects[355,356] (Fig. 6b).

In summary, successful translation of galectin biology into effective therapies will require the integration of mechanistic insights with advanced drug design, single-cell, spatial, and glycoengineering technologies. Such multidisciplinary approaches are essential for overcoming current translational barriers, enabling context-specific targeting of galectins, and ultimately advancing precision medicine across a broad range of diseases.

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AUTHOR CONTRIBUTIONS
A.M.P., A.S., and A.J.-A. conceived and designed the manuscript. A.M.P. and A.S. wrote the first draft of the manuscript, generated the figures, compiled the references, and conducted revisions. A.J.-A. contributed to writing the manuscript, edited the manuscript, and provided supervision and funding. All authors have read and approved the article.

ADDITIONAL INFORMATION
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